



SOFTWARE REQUIREMENT SPECIFICATION

Kawai Resort & Tour Hub

Group 2

SE2003-NET

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I. Record of Changes

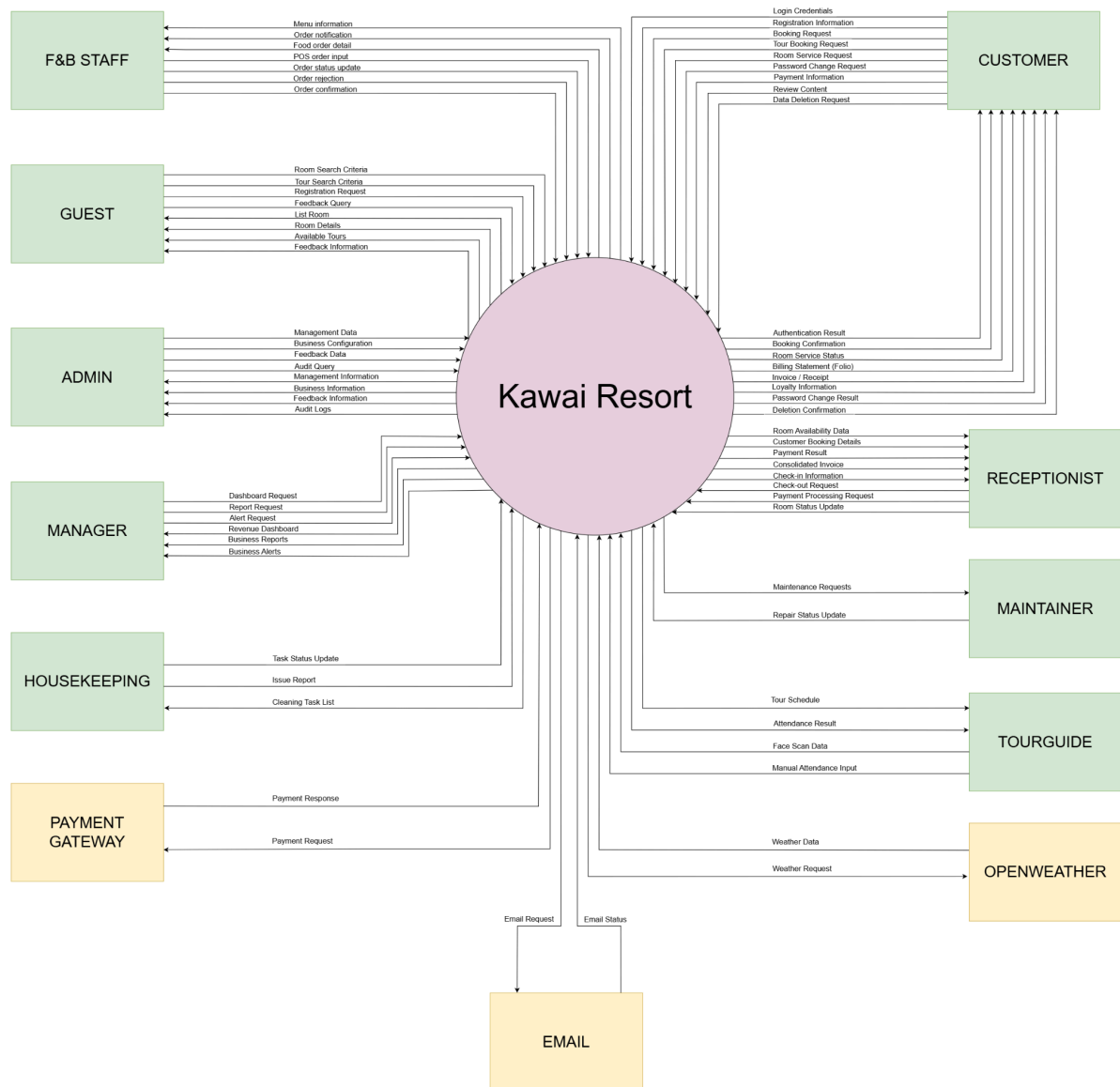
Date	A* M, D	In charge	Change Description
24/5/2026	A	Team	Added Context Diagram, User Requirement (Actors), User Case Diagram, Entity Relationship Diagram
25/5/2026	A	Team	Added System Functionalities
27/5/2026	A	Team	Added User Requirement (Use Cases)
28/5/2026	A	Team	Added Entities Description
30/5/2026	M	Team	Updated Use Case Diagram
1/6/2026	M	Team	Updated Use Case Diagram and Screens
2/6/2026	M	Team	Updated Context Diagram and Use Case Diagram
2/6/2026	A	Team	Added Use Case Specifications
3/6/2026	A		Added Main Business Processes, UC7-15, Business Rules
3/6/2026	D		Deleted Entity Relationship Diagram

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Overall Requirements

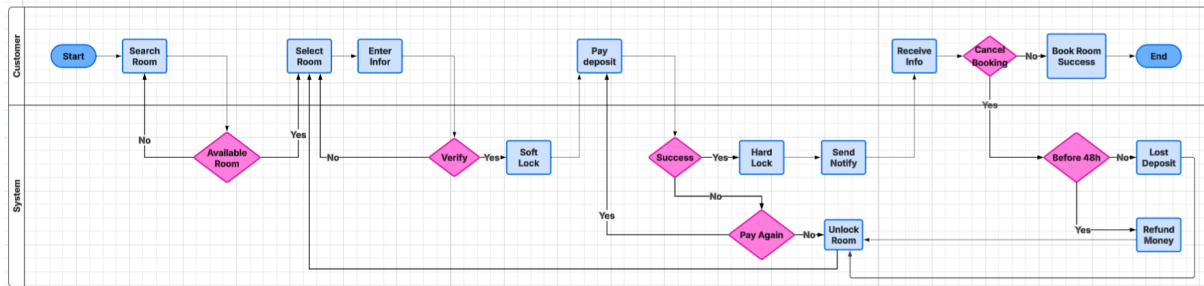
1.1 Context Diagram



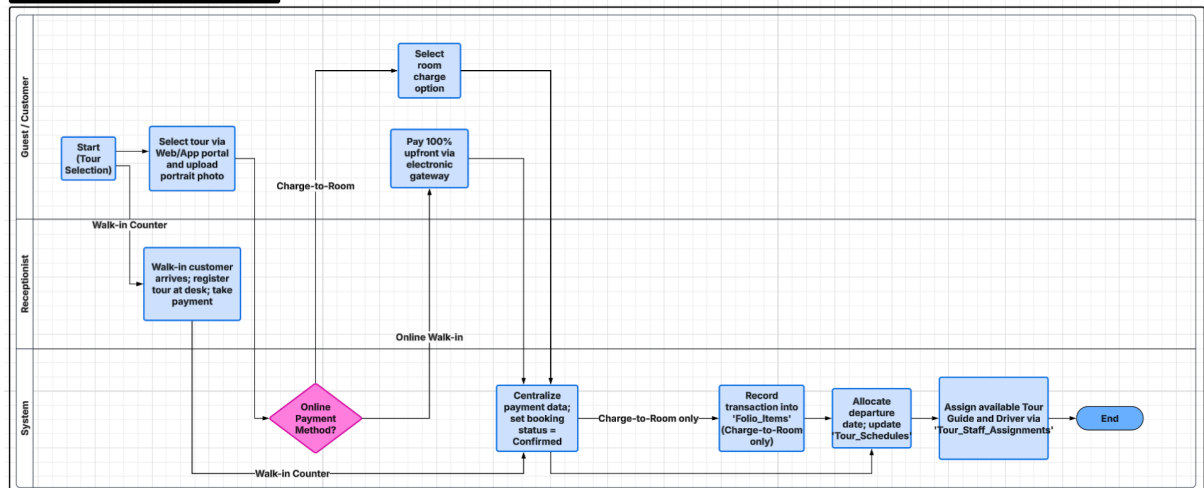
URL: [Context Diagram](#)

1.2 Main Business Processes

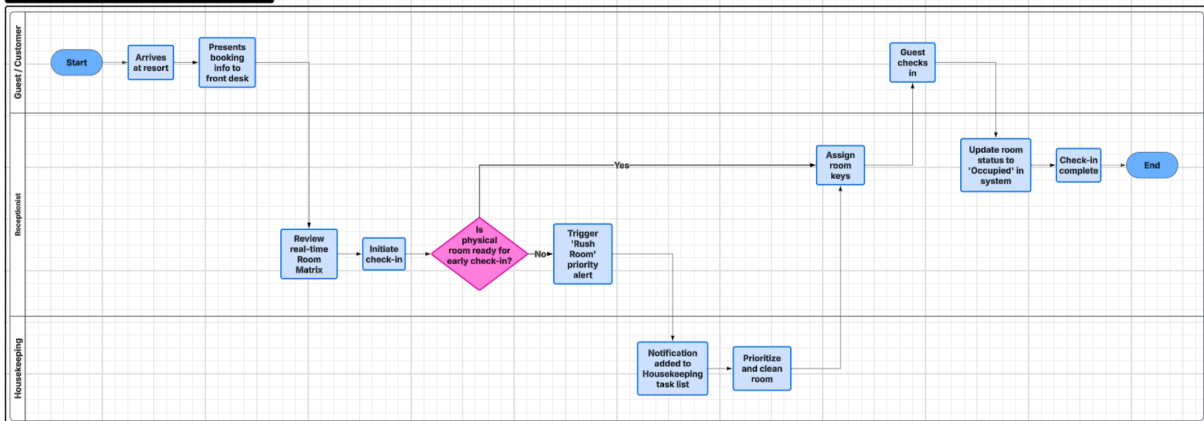
1.2.1 BookRoom



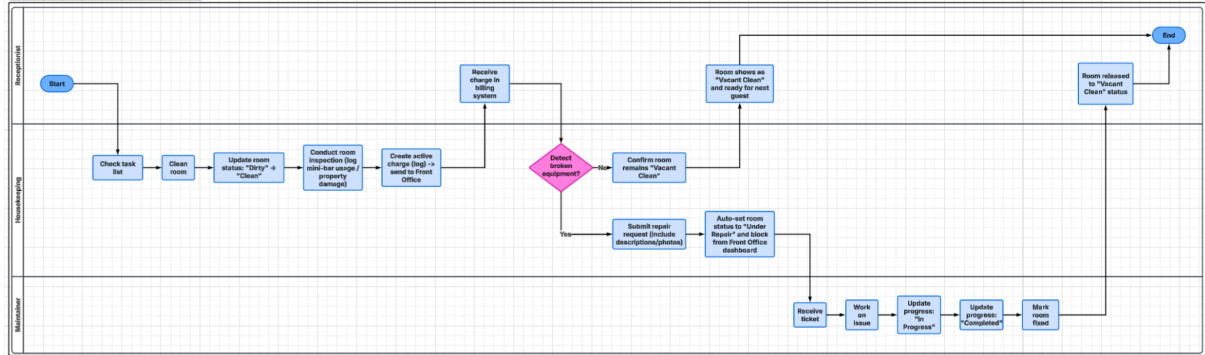
Tour Booking & Staff Allocation Workflow (Kawal Resort)



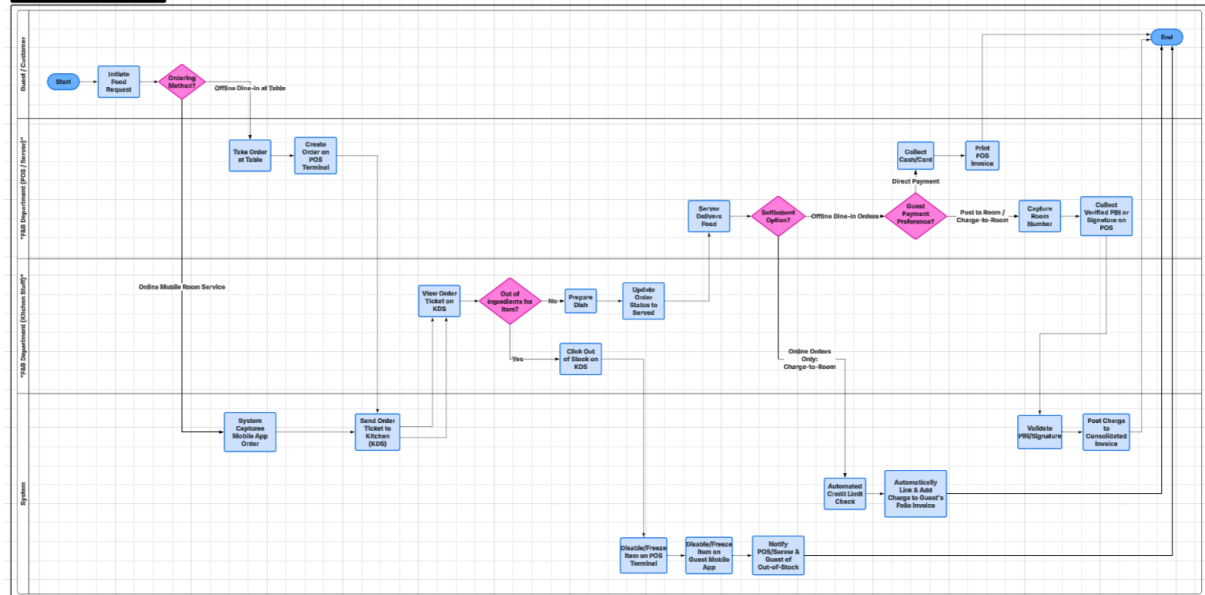
Front Office Check-in & Coordination Workflow (Kawal Resort)



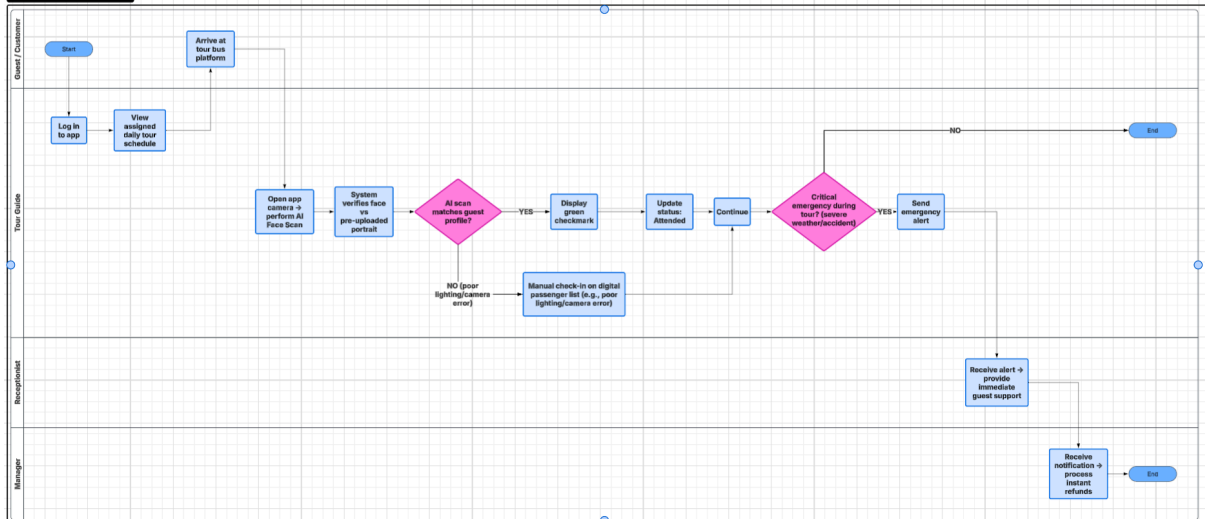
Housekeeping & Maintenance Operation Workflow

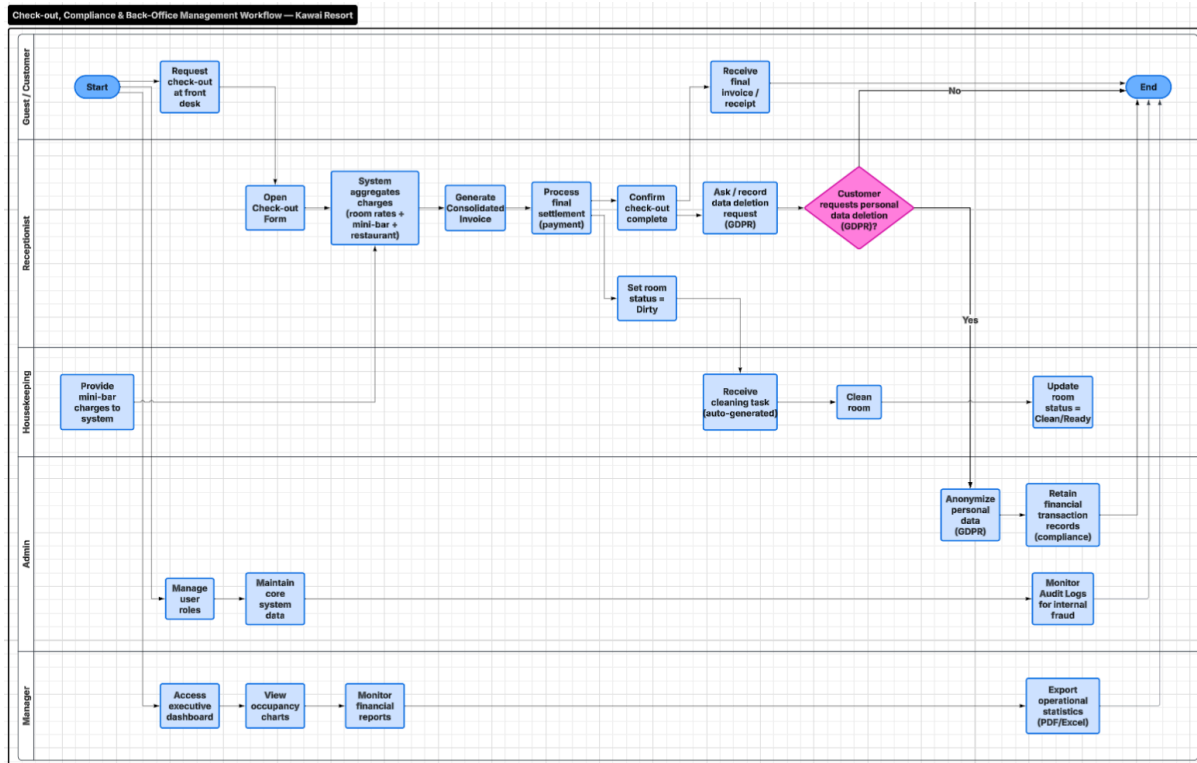


F&B Restaurant POS & Kitchen System Workflow



Tour Ops & AI Face Scan Workflow





Step #	Step Name	Detailed Description	Role	Note
1	Search Available Rooms	1. Activity: Guest filters room categories by check-in/out dates, guest count, and requirements. System validates availability and calculates real-time pricing. 2. Input: Search criteria (dates, capacity, category). 3. Output: List of available room categories with descriptions and base prices.	Guest / Customer	Searching can be performed without authentication.

2	Book Room & Pay Deposit	1. Activity: Customer selects rooms, confirms stay details, and initiates payment. System processes deposit through VNPay Sandbox to secure the booking. 2. Input: Booking details, guest information, selected rooms. 3. Output: Confirmed booking record, deposit transaction receipt, booking status updated to Confirmed.	Customer	If deposit payment is not received within 15 minutes, the booking is automatically cancelled (BR-FO-02)
3	Check-in & Allocation	1. Activity: Receptionist verifies guest identity (ID/Passport), confirms payment, assigns a physical room, and configures spending authorization (Credit Limit/PIN). 2. Input: Booking reference, identification documents. 3. Output: Assigned room number, updated Room/Booking status to Checked_In, active guest folio.	Receptionist	Primary guest must be at least 18 years old (BR-FO-03).

4	Check-out & Final Invoice	1. Activity: Receptionist reviews all folio charges (room, F&B, tours), applies any promotion codes, processes final payment, and issues a consolidated invoice. 2. Input: Outstanding folio items, final payment method. 3. Output: Consolidated invoice, payment confirmation, room status set to Vacant_Dirty, automatic trigger of cleaning task.	Receptionist	Checkout is prohibited unless all outstanding balances are settled (BR-FIN-01).
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1.3 User Requirements

1.3.1 Actors

#	Actor	Description
1	Administrator	System administrator. Manages system configurations, user accounts, and monitors overall system operations
2	Guest	Unauthenticated user. Can browse system information, view room details, and check availability, but cannot book rooms or access personalized features.
3	Customer	Registered and authenticated user. Can browse information, book rooms, tours, and services, manage booking history, leave reviews, and update their personal profile.
4	F&B Staff	Hotel Food & Beverage staff. Manages food and beverage orders, processes payments, and updates order statuses for restaurant or room service requests.
5	Receptionist	Front desk staff. Responsible for managing guest reservations, handling check-in and check-out processes, assigning rooms, and generating operational reports.

6	HouseKeeping	Room attendant and cleaning staff. Responsible for viewing assigned cleaning tasks, updating room cleanliness statuses, and reporting maintenance maintenance issues.
7	Tour Guide	Tour leader or guide. Responsible for viewing assigned tour schedules, itineraries, and managing guest attendance for the tours.
8	Maintainer	Maintainer and technical staff. Responsible for viewing reported facility maintenance requests and updating the status of technical issues after fixed.

1.3.2 Use Cases (UC)

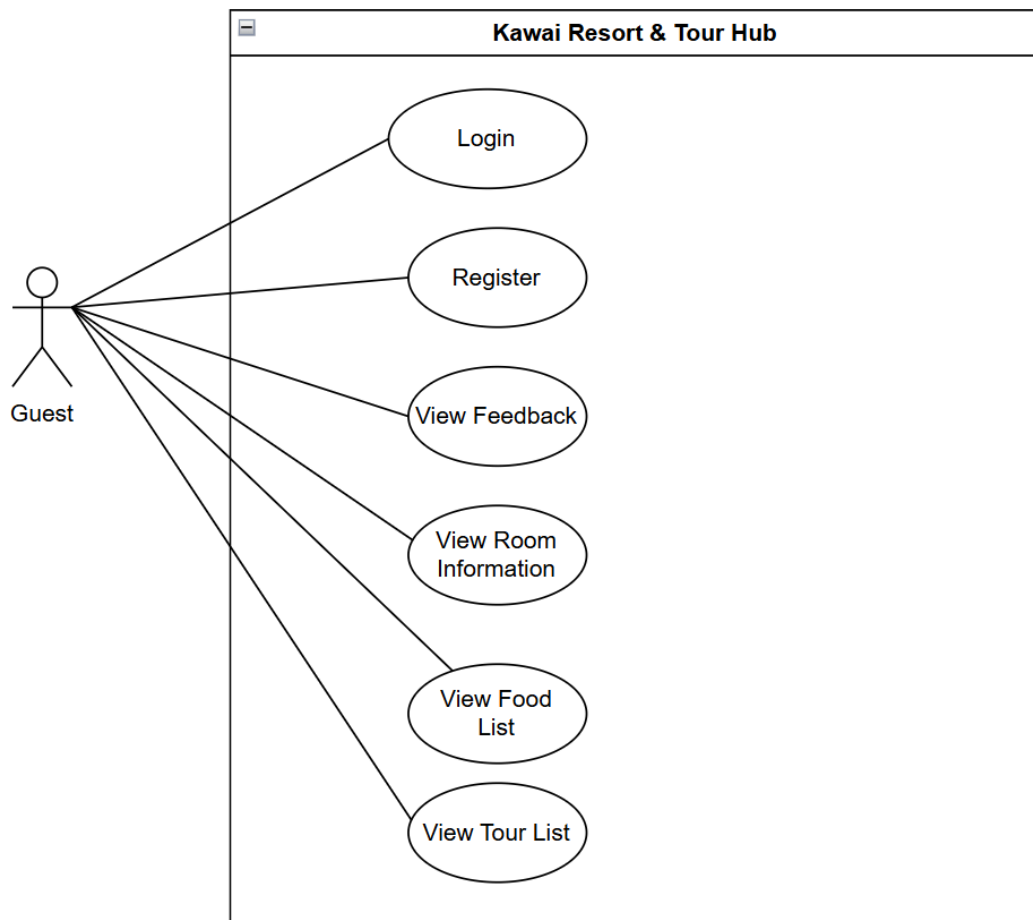
ID	Use Case	Feature	Use Case Description
01	Login System	Authentication	All users (Administrator, Manager, F&B Staff,...) log into the system using their assigned credentials.
02	Register Account	Authentication	Allows public guests to create a new customer account on the website portal.
03	Verify Authentication via OTP	Authentication	Enhances security by requiring an OTP or Email Code verification during login or key actions.
04	Reset Password	Authentication	Allows users who forgot their password to safely request a reset link or code via registered email.
05	Manage Profiles & Upload Identity Documents	User Profiles	Customers manage personal profiles and upload encrypted photos of National ID/Passport for legal compliance.
06	Upgrade Dependent to Customer	User Profiles	Receptionists upgrade checked-in dependents (≥ 18 years old) to independent Customer accounts without losing history.
07	Request Personal Data Deletion	Data Protection	Allows customers to exercise their "Right to Deletion" by requesting data anonymization after checking out.
08	Manage Staff Accounts & Roles	Master Data	Administrators create, lock, or update employee accounts and dynamically assign granular operational roles.
09	Manage Master Data	Master Data	Administrators perform CRUD operations on room categories, public restaurant menus, and tour catalogs.
10	Search Available Rooms	Room Booking	Customers filter and search for vacant rooms based on selected date range, guest count, and room type.
11	Book Room & Pay Online Deposit	Room Booking	Customers book multiple rooms in one order and securely process a deposit payment via VNPay Sandbox.
12	View Expected Arrivals Matrix	Front Desk	Receptionists monitor a daily visual dashboard of room matrices, expected arrivals, and pending checkout schedules.
13	Check-In & Allocate Physical Rooms	Front Desk	Receptionists execute check-in, assign specific physical room numbers, and collect temporary residence data.

14	Authorize Spend Limit	Front Desk	Receptionists enable sub-rooms to charge expenses directly to the master room up to a strict maximum credit limit.
15	Order Room Service	Restaurant POS	Checked-in customers browse the in-app dining menu and place food orders to be delivered directly to their room.
16	Order Dine-In Meal	Restaurant POS	F&B Staff log into restaurant POS terminals to place food and beverage orders for customers dining at tables.
17	Track Kitchen Order Ticket (KOT)	Restaurant POS	Kitchen staff view incoming food orders and track state lifecycle updates (Pending → Preparing → Served).
18	Flag Out-of-Stock Dishes	Restaurant POS	Kitchen staff mark exhausted dishes as out-of-stock, automatically greying them out on all POS terminals and customer interfaces.
19	Reserve Dining Table	Restaurant POS	Customers or staff reserve specific restaurant dining tables by setting dates, target times, and total party sizes.
20	Charge Meal Bill to Room	Hospitality Finance	F&B Staff route dining expenses directly into the guest's folio to defer payment until the final checkout.
21	Schedule Tour Deliveries	Tour Booking	Tour managers set up practical daily vehicle departures (Tour_Schedules) from parent packages, assigning guides and seats.
22	Browse & Book Tours	Tour Booking	Customers browse tour itineraries with live OpenWeather integrations and book tickets for explicit participant counts.
23	Auto-Cancel Under-Capacity Tours	Tour Booking	The system triggers automated cancellations and issues 100% refunds if minimum occupancy isn't met 24 hours prior.
24	Execute AI Face Scan Attendance	Tour Booking	Tour guides open mobile cameras to scan passenger faces, matching identity vectors real-time to log attendance.
25	Submit 5-Star Reviews	Feedback System	Customers write detailed feedback and submit structured 5-star ratings within 7 days of checking out or completing a tour.
26	Moderate Customer Reviews	Feedback System	Administrators moderate (hide or display) guest reviews on the public web portal while saving clear audit trails.
27	Track Itemized Room Debt (Folio)	Hospitality Finance	Customers monitor per-room itemized expenses (Folio_Items) in real-time on their personal screens.
28	Generate Consolidated Invoice	Hospitality Finance	The system aggregates all accommodation, unpaid dining, and tour charges into one final combined invoice.
29	Process Final Checkout Payment	Hospitality Finance	Receptionists record final digital or cash payments, mark invoices as settled, and print digital receipts.
30	Run Night Audit & Daily Closing	Hospitality Finance	Managers trigger a "Daily Closing" function to update occupancy statistics, balance logs, and roll over business dates.

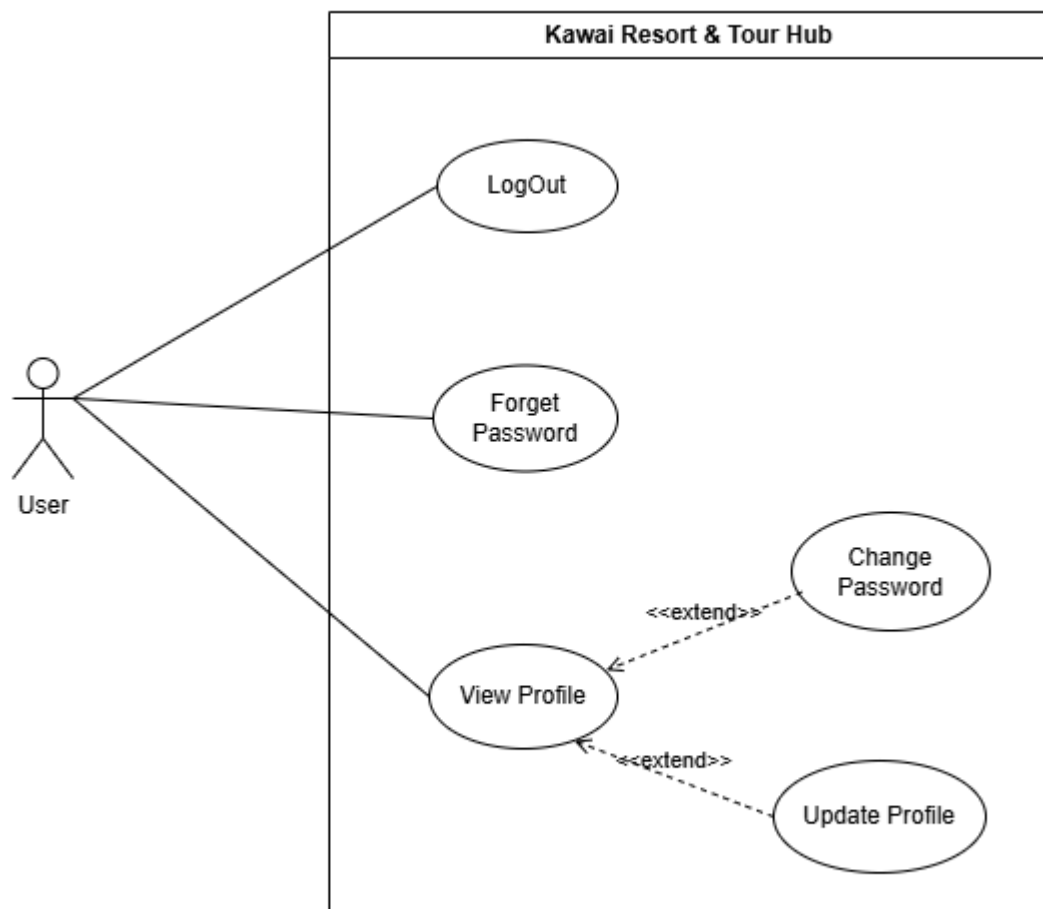
31	Monitor Financial Dashboards	Hospitality Finance	Executive managers view interactive dashboards splitting revenues into Room, F&B, and Tour sectors per USALI standards.
32	Track Anti-Fraud Audit Logs	Master Data	Administrators audit back-end historical traces (Audit_Logs) by filtering rooms or timestamps to trap internal fraud.
33	Export Financial Reports	Hospitality Finance	Managers generate and download comprehensive monthly operation reports in structured Excel or PDF formats.
34	Auto-Trigger Checkout Cleaning	Housekeeping	Completing a room checkout automatically posts a Checkout_Clean mobile task and shifts room states to Vacant_Dirty.
35	Update Cleaning Status	Housekeeping	Cleaners mark cleaning tasks as complete on mobile, triggering a state shift back to Vacant_Clean on the front-desk matrix.
36	Log Maintenance Requests	Maintenance	Housekeepers or staff report broken room appliances on-site, moving the room state to Maintenance to close inventory.
37	Resolve Engineering Failures	Maintenance	Technicians receive mobile maintenance repair tickets and input diagnostic summaries to unlock rooms back into service.
38	Apply Promotion Campaign Vouchers	Financial Controls	Receptionists select registered promotional campaign codes from centralized tables instead of entering manual bill discounts.
39	Log Multi-Stage Transactions	Financial Controls	The system tracks chronological financial movements including advances, split installments, final settlements, or refunds.
40	Set Dynamic Holiday Pricing	Financial Controls	Managers adjust pricing algorithms dynamically based on specific holiday schedules or custom calendar dates.

1.3.2 Use Case Diagrams

1.3.2.1 UCs for Guest

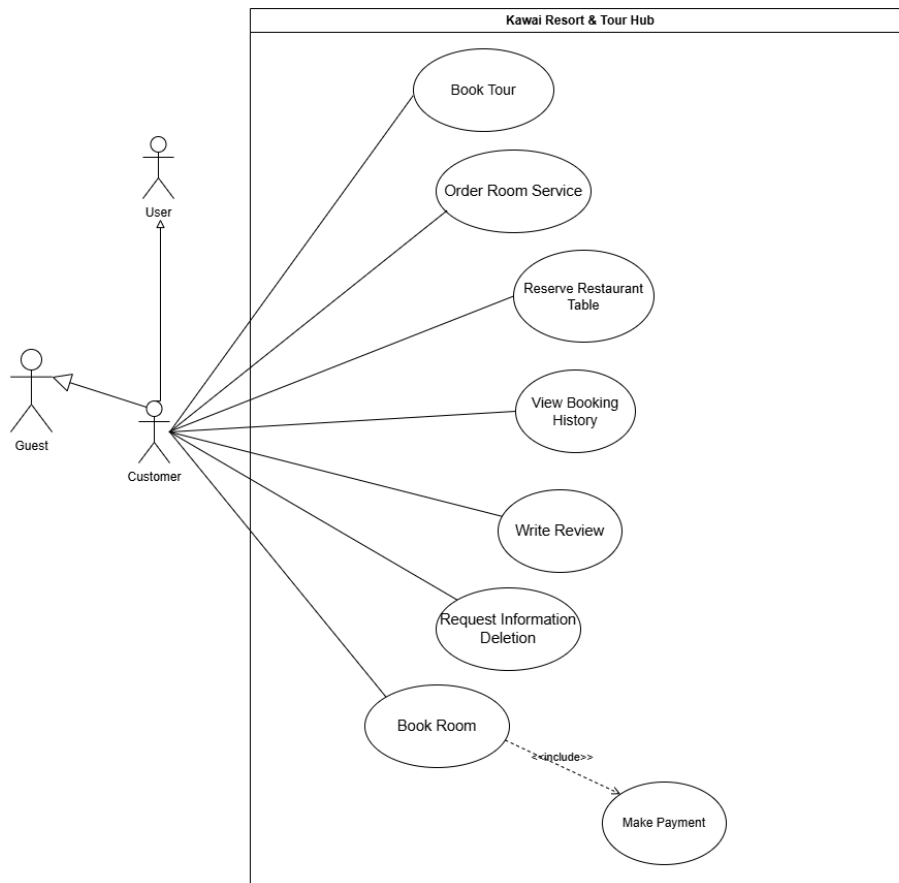


>> [Guest Usecase](#)



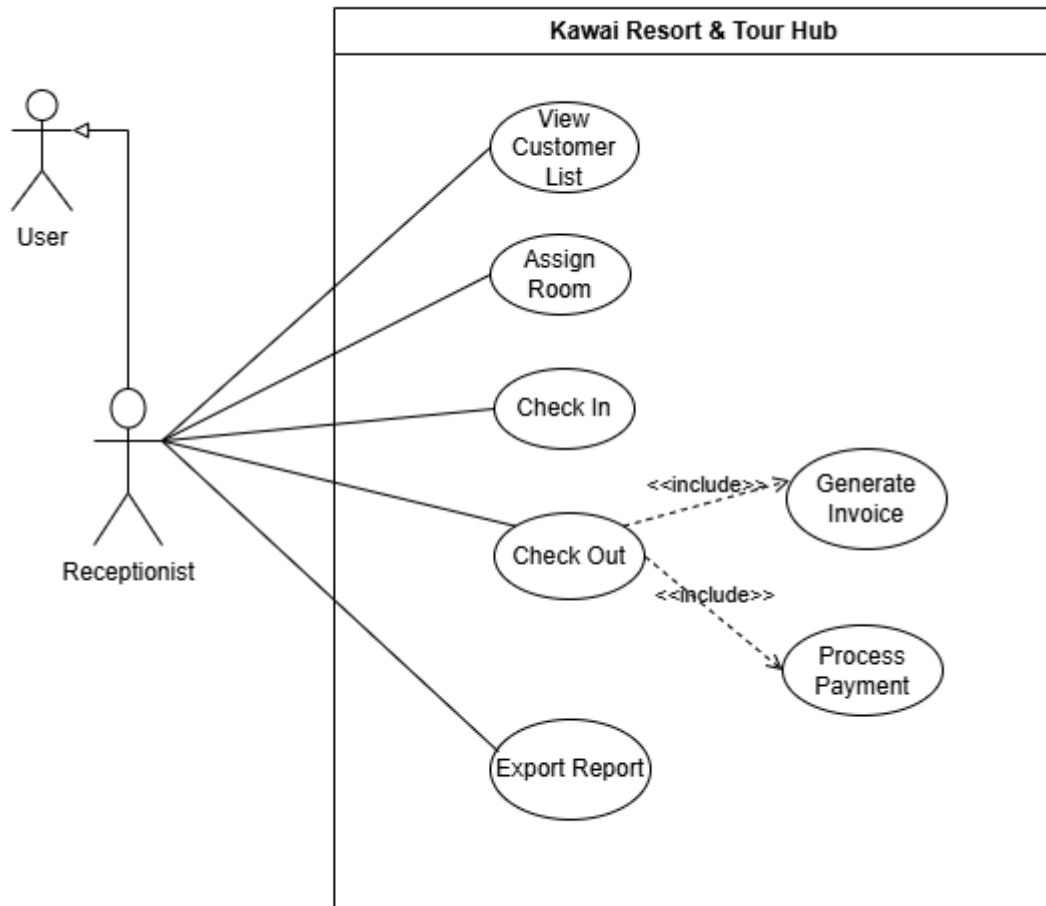
⇒ [User](#)

1.3.2.2 UCs for Customer



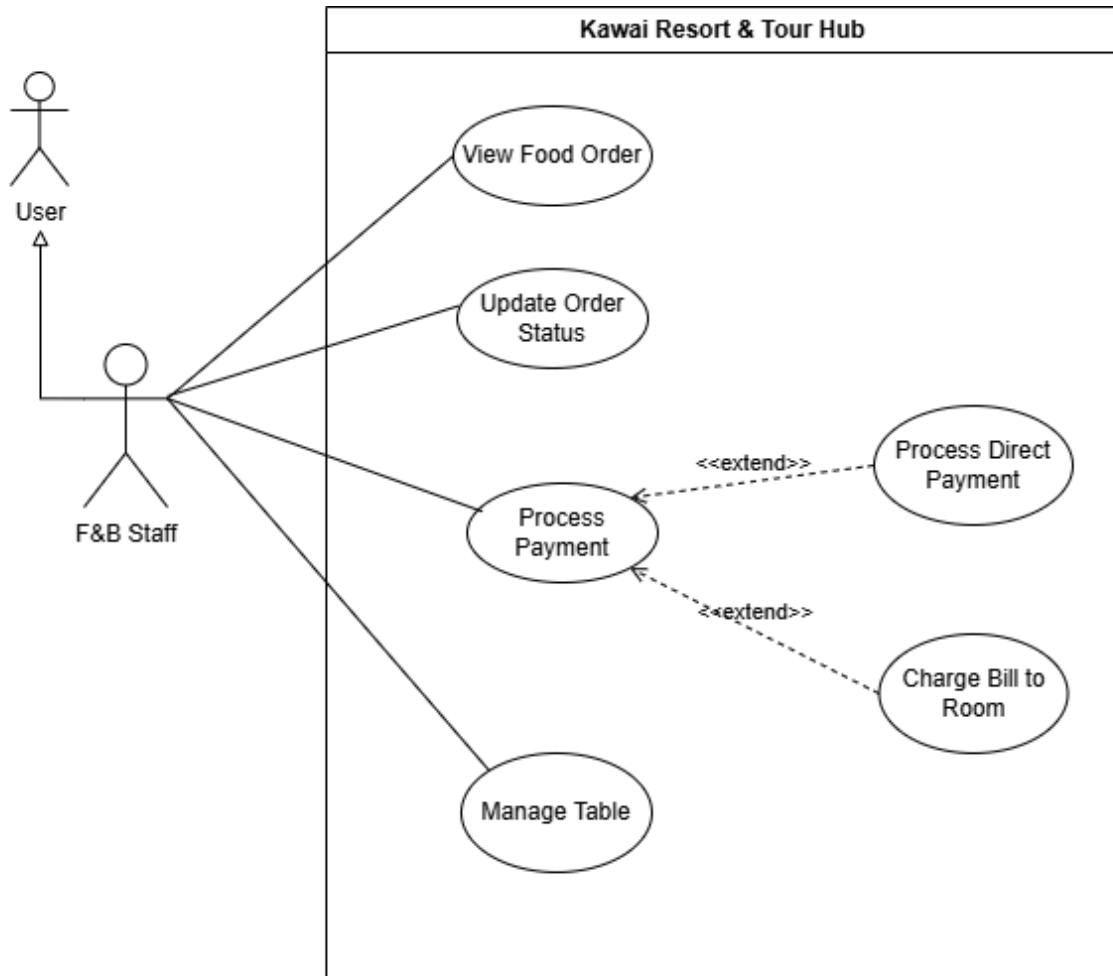
>> [Customer-UseCase](#)

1.3.2.3 UCs for Receptionist



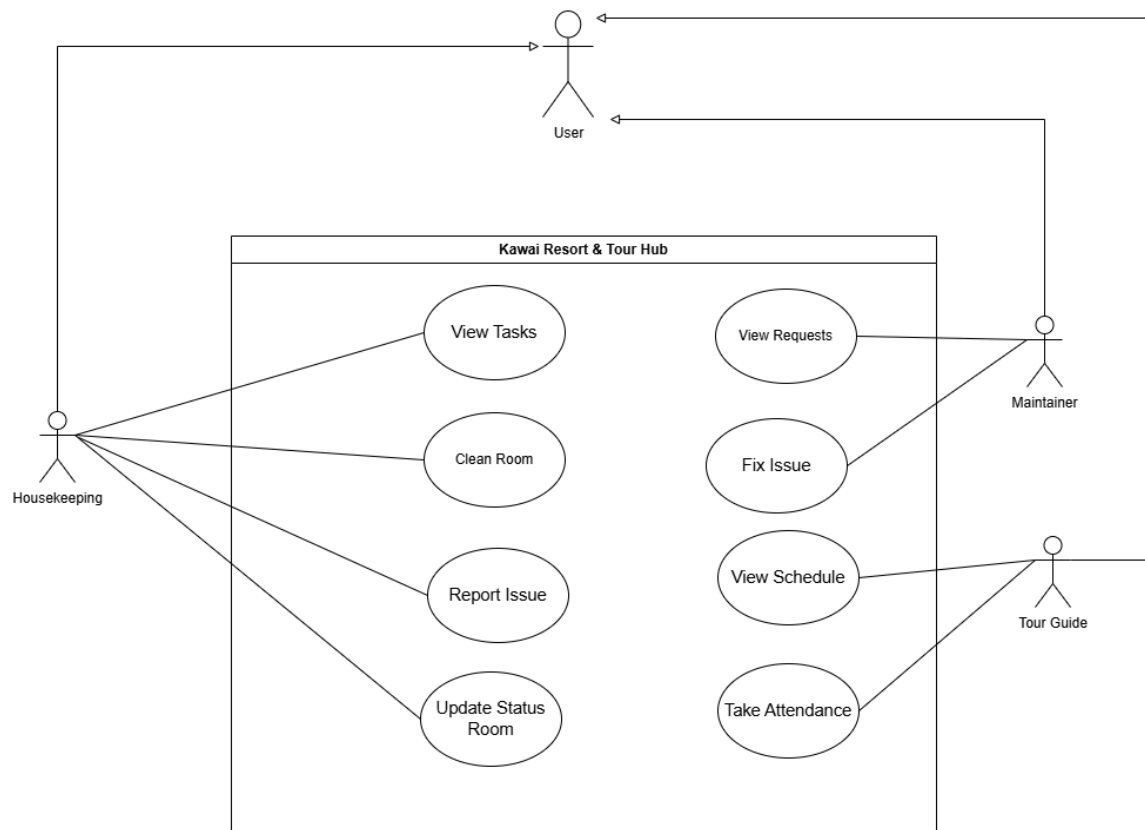
>> [Receptionist_UseCase](#)

1.3.2.4 UCs for F&B Staff



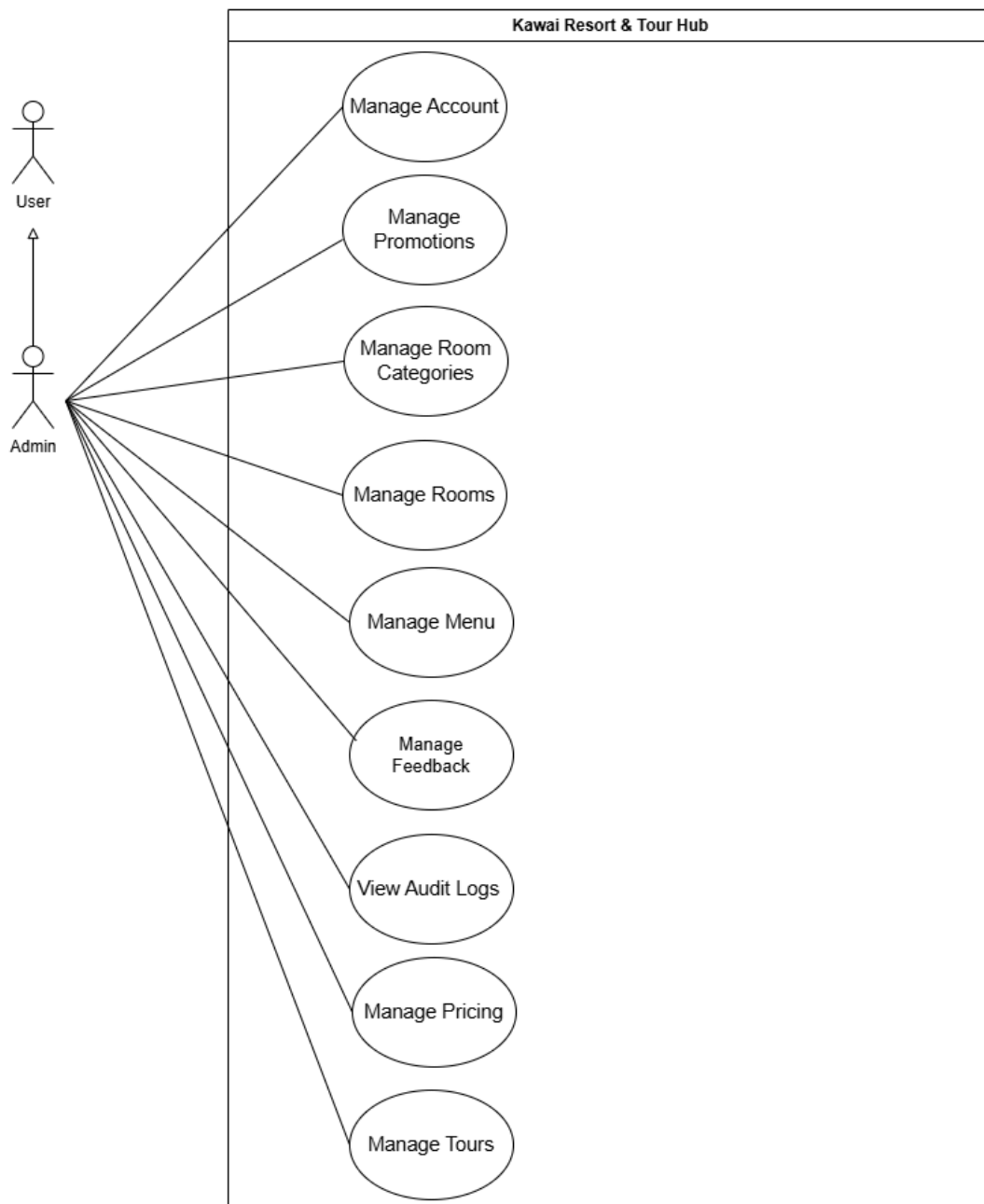
>> [F&B Staff](#)

1.3.2.5 UCs for Tour Guide, Housekeeping, Maintenance



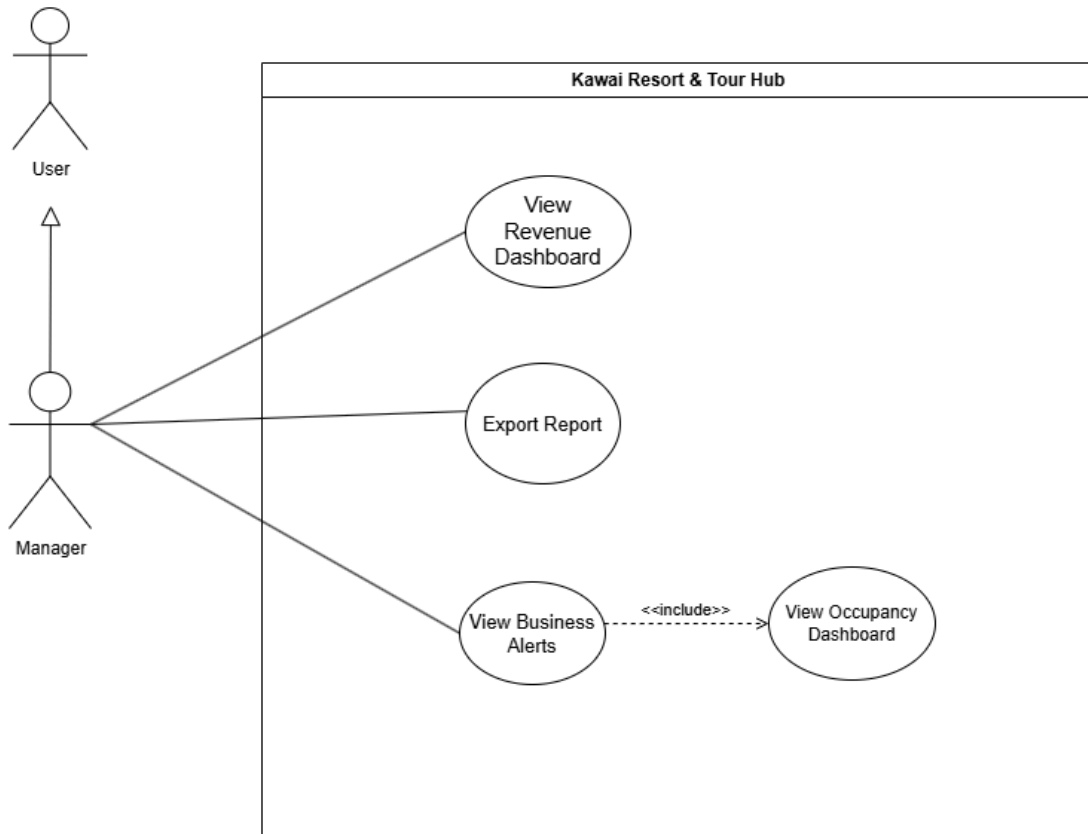
>> [HouseKeeping_Usecase](#)

1.3.2.6 UCs for Admin



>> [Admin_Usecase](#)

1.3.2.7 UCs for Manager



=>> [Manager](#)

1.4 System Functionalities

1.4.1 Screens Flow

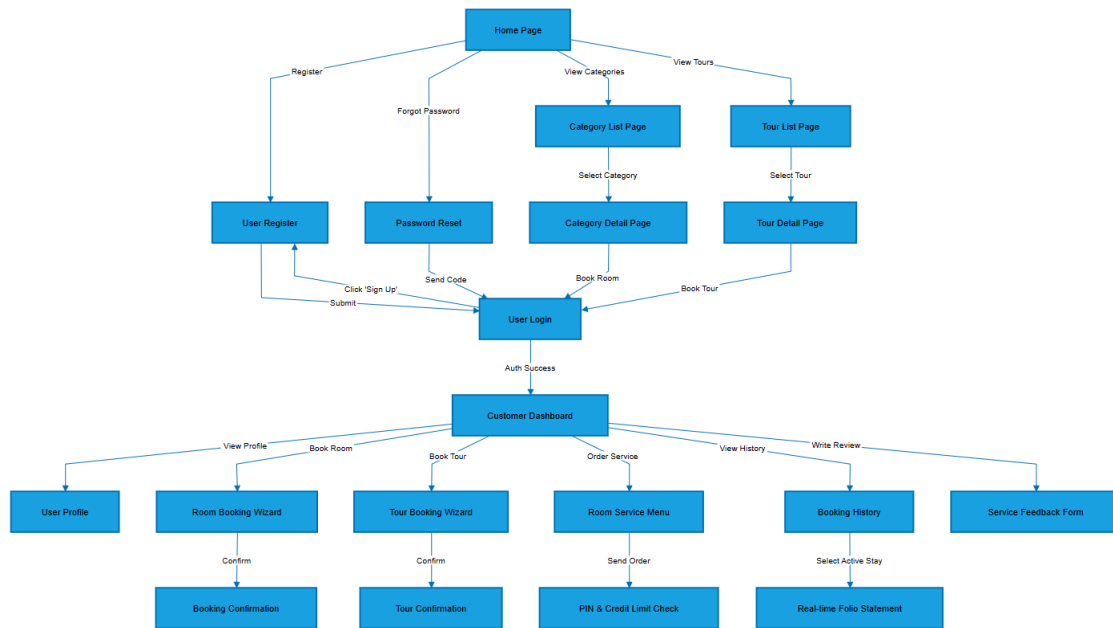


Figure 1: UserScreen Screen Flow

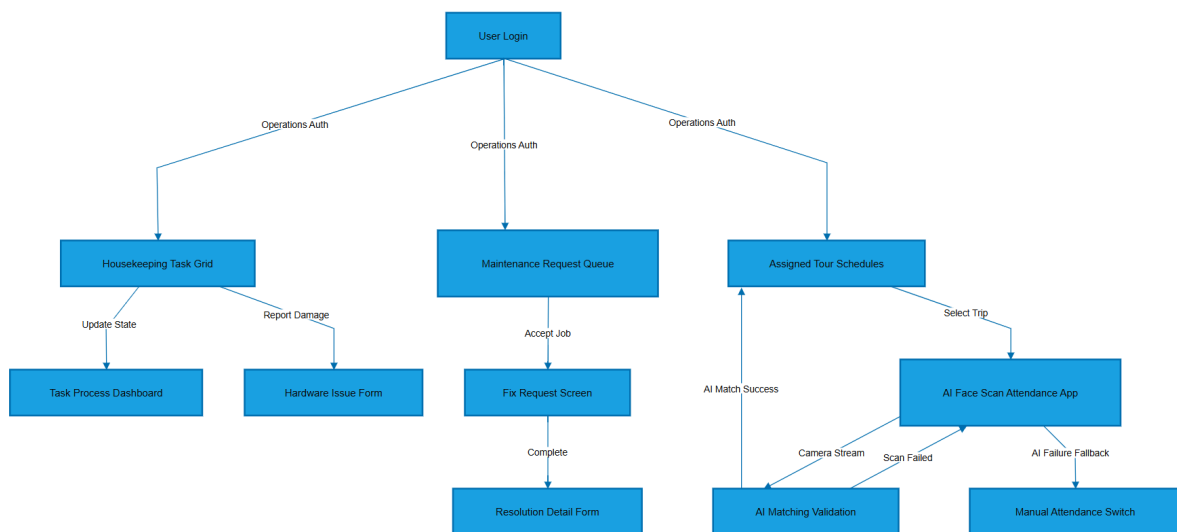


Figure 2: Housekeeping&Maintainer Screen Flow



Figure 3: Receptionist Screen Flow

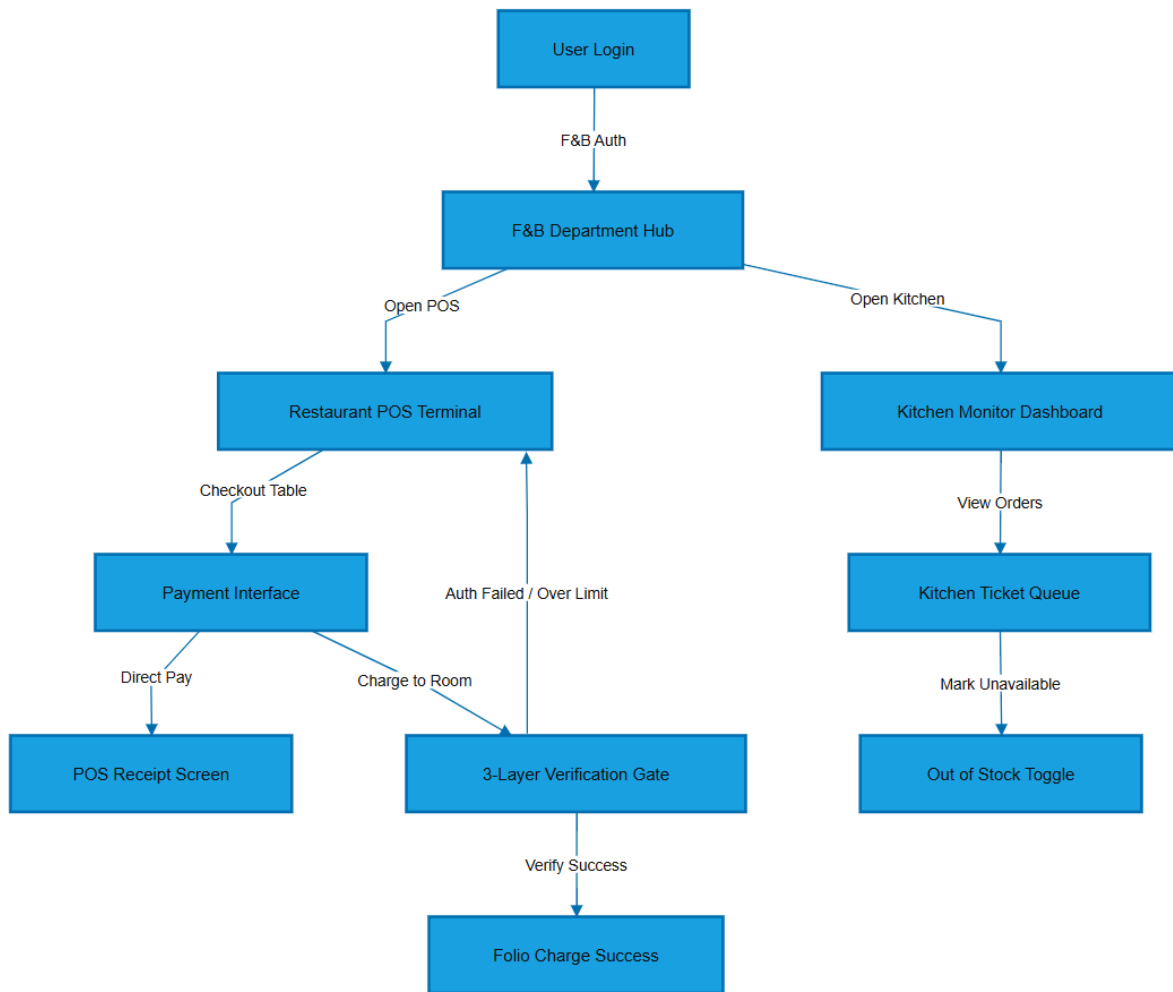


Figure 4: F&B Screen Flow

Figure 5: Manager Screen Flow

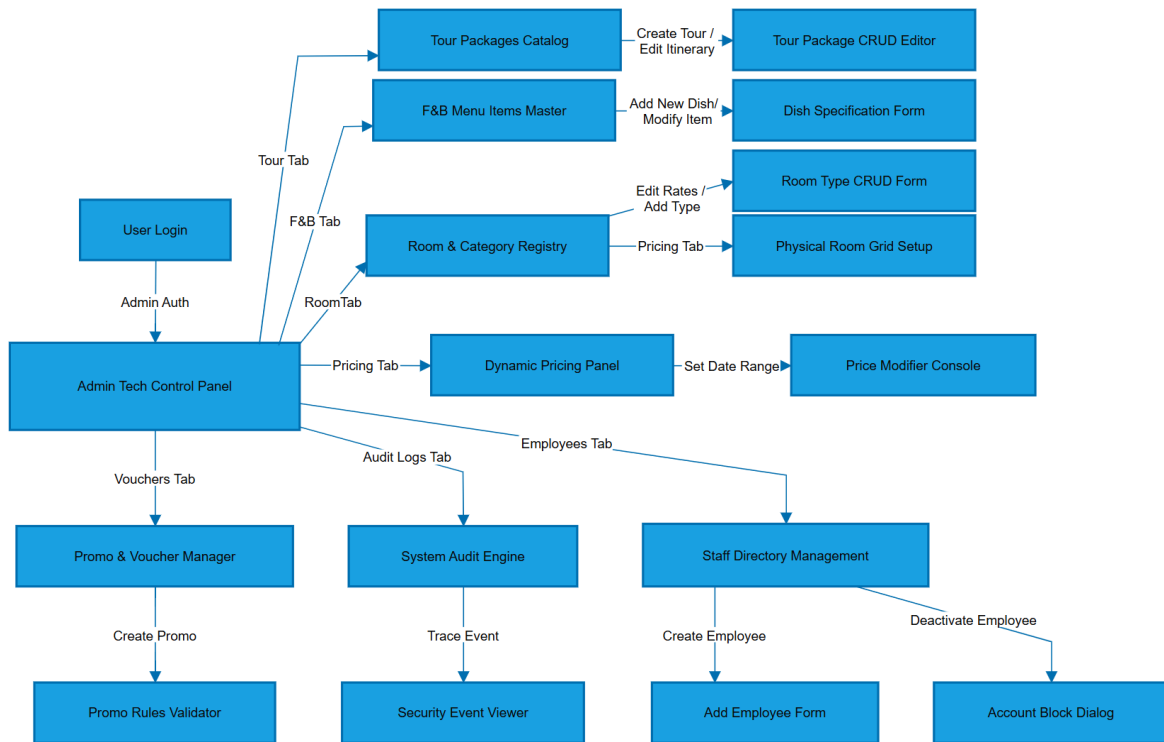


Figure 6: Admin Screen Flow

1.4.2 Screen Authorization

Screen	Guest	Customer	Receptionist	F&B Staff	Kitchen Staff	Tour Guide	Housekeeping	Maintenance	Admin/ Manager
Single Login Portal	X	X	X	X	X	X	X	X	X
Public Home Page	X								
User Registration Form	X								
Member Dashboard		X							
Room Service Menu		X		X					
Booking History & Folio		X							
Service Feedback Form		X							
Room Matrix Dashboard			X						
Check-in Processing Form			X						

Checkout & Bill Workspace			X						
Restaurant POS Terminal				X					X
Payment Verification Gate				X					
Kitchen Status Dashboard					X				X
Assigned Tour Schedules						X			
AI Face Scan Attendance						X			
Housekeeping Task Grid							X		
Maintenance Request Queue								X	
USALI Financial Analytics									X
Config & Audit Log Panel									X
Staff Directory Management									X

1.4.3 Non-UI Functions

#	Feature	System Function	Description
1	Authentication & Security	Password_Encryption_Filter	Extracts and encrypts raw user passwords using the BCrypt hashing algorithm before performing data matching in the Accounts table.
2	Authentication & Security	Role_Based_Routing_Engine	A Spring Security filter that analyzes the role_name of the authenticated Token to automatically configure and trigger URL redirection for each Actor.
3	Room Booking	Pending_Booking_Auto_Cancellation	Runs a background Task Scheduler every 5 minutes to automatically scan and cancel room reservations (status = 'Cancelled') if the 30% deposit is not completed within 15 minutes.
4	Front-Desk Operations	Dependent_Account_Auto_Generation	Triggered when the Receptionist upgrades a dependent's account, automatically executing parallel data insertion into both the Accounts and Customers tables.

5	Front-Desk Operations	Housekeeping_Task_Trigger	A database Trigger that detects when a room booking status changes to Checked_Out and immediately initializes a new cleaning task entry in the Housekeeping_Tasks table.
6	F&B Billing	Credit_Limit_Realtime_Validator	A Constraint Validator that executes in the background when a guest orders a service, aggregating the current outstanding folio balance with the new order total to ensure it does not exceed the room's Credit_Limit.
7	F&B Billing	Kitchen_Notification_Dispatcher	Uses WebSockets or a Message Broker to automatically dispatch Kitchen Order Tickets (KOT) to the kitchen monitor as soon as a room service order passes PIN verification.
8	Kitchen Inventory	Menu_Item_Availability_Sync	Triggered when the kitchen marks an item as out of stock, updating the database state to is_available = false and broadcasting real-time UI updates to all POS terminals and guest devices.

9	Tour Attendance	AI_Face_Vector_Matching_Service	A background Python AI service that receives the camera video stream, extracts facial feature vectors, and compares them against the encodings.pickle template file to verify identity.
10	Tour Attendance	Attendance_Status_Synchronizer	Receives positive match results from the AI Service or manual inputs from the Tour Guide to execute the UPDATE query, changing attendance_status to 'Attended'.
11	Housekeeping & Maintenance	Room_Status_State_Machine	Automatically manages and syncs the physical room state (Rooms table) on the Receptionist's Matrix Dashboard whenever a housekeeping or maintenance staff member updates task progress on mobile.
12	Housekeeping & Maintenance	Maintenance_Request_Auto_Bridge	Triggered when housekeeping reports equipment damage, automatically inserting a new issue record into the Maintenance_Requests table and switching the room status to 'Maintenance'.

13	Financial Accounting	USALI_Revenue_Decomposer	Automatically decomposes the gross revenue from finalized invoices, calculating and allocating financial data into three distinct departmental revenue categories (Rooms, F&B, Tours) following USALI standards.
14	Marketing & Promotions	Voucher_Validity_Checker	Validates promotional codes against predefined constraints (expiration date, global usage limit, and minimum order amount) before applying discounts to the total invoice.
15	Security Auditing	AOP_Data_Interception_Logger	Utilizes Spring AOP (Aspect-Oriented Programming) to transparently intercept modifications to financial tables, capturing old/new values, staff ID, and system logs into the Audit_Logs table.
16	Staff Directory	Employee_Account_Transactional_Link	Encapsulates the staff onboarding process within a @Transactional block, ensuring a complete rollback of both employee records and login accounts if any insertion error occurs.

URL: [ERD](#)

#	Entity	Description
1	Account	Stores account information for all system users (including Customers, Staff, and Admins), such as usernames, encrypted passwords, emails, phone numbers, and account statuses.
2	Role	Manages system roles and permissions (e.g., Admin, Staff, Customer,) to restrict and authorize access to specific features.
3	Employees	Stores detailed personal and professional profiles of the staff members linked to an account, managing information such as full name, gender, national ID (CCCD), phone number, email, and salary details.
4	Audit_Logs	Records security and data modification history within the system. It tracks which employee performed what action (e.g., Insert, Update, Delete) on a specific table, keeping old versus new values alongside IP addresses and timestamps for security auditing.
5	Hotel_Operations	Manages daily internal hotel tasks and operational workflows (such as room cleaning, maintenance, or room service). It links tasks to specific hotel rooms, assigns them to an operational staff member (staff_id), designates a supervisor, and tracks priority, task status, and execution timelines.

6	Tour_Staff_Assignments	Handles the staffing and assignment of employees to specific tour schedules. It maps which employee is assigned to a tour schedule and defines their specific operational role during that tour (e.g., Tour Guide, Driver, Coordinator).
7	Dynamic_Pricing	Manages seasonal or time-based price adjustments for hotel room categories. It stores specific date ranges (start_date to end_date) and a pricing multiplier or modifier to calculate flexible room rates dynamically during holidays, weekends, or peak seasons.
8	Room_Categories	Defines the types or classifications of hotel rooms available (e.g., Standard, Deluxe, Suite). It captures the structural details of each category, including the category name, standard baseline price (base_price), and maximum guest accommodation capacity.
9	Rooms	Stores the physical inventory of individual hotel rooms. Each record tracks a specific room identifier (room_number), references its operational classification (category_id), monitors its availability state (room_status), and keeps track of any ongoing reservation via the active booking detail.
10	Room_Bookings	Extends the global booking entity specifically for hotel room stays. It acts as a 1:1 sub-type mapping to booking_id (PK, FK) and captures financial policy controls like deposit_amount, credit_limit, check-in/out schedules, strict cancellation deadlines, and a secured personal_pin_hash for digital room access or self-check-in verification.
11	Bookings	Acts as the central, high-level transaction record for any reservation made by a customer. It stores core structural metadata including the booking date, total accumulated price (total_price), current global state (booking_status), source of the booking (e.g., Online, Walk-in), and data concurrency control (version).
12	Tours	Manages the catalog of holiday packages and sightseeing trips offered by the business. It stores core details such as the official tour name, the standard base cost calculated per individual passenger (price_per_person), and the maximum group size constraint (max_capacity).
13	Tour_Schedules	Handles the specific operational departures and calendar events for the scheduled tours. It charts exactly when a tour will run by defining its precise departure date, departure time, and current scheduling status (e.g., Active, Fully Booked, Departed, Cancelled).
14	Customers	Stores comprehensive personal profiles and loyalty data for registered guests. It tracks details such as full name, gender, identification (cccd, passport), contact information (phone, email), accumulated reward points (loyalty_points), and membership tier levels.
15	Dependents	Manages information about family members or companions traveling alongside a primary customer. It links directly to a

		customer record and tracks the dependent's name, birth date, and legal ID or passport number for check-in compliance.
16	Room_Booking_Details	Actively logs the specific operational segments of a room reservation. It links the primary booking reference to an allocated physical room (room_id), its category, and dynamically records the calculated room_charge and current status (detail_status). It also references a guest_customer_id if the room occupant differs from the booking owner.
17	Table_Reservations	Manages dining reservations at the hotel's internal restaurant facilities. It maps a customer to a specific physical table assignment (table_id) and tracks the intended reservation date (reserve_date), time, and fulfillment status.
18	Restaurant_Tables	Stores the physical inventory of dining tables within the hotel's internal restaurant facilities. It tracks specific data for each table, including its official identifier (table_number) and maximum seating accommodation (capacity).
19	Reviews	Stores quality assessments, service ratings, and text feedback written by customers. It enables granular feedback loops by allowing guests to rate either a specific room stay (room_booking_detail_id) or a specific sightseeing excursion (tour_booking_id).
20	Tour_Bookings	Extends the global booking entity for sightseeing excursions. It maps the parent booking_id directly to a specific operational trip (schedule_id), keeping track of the total guest headcount (participant_count) and their overall attendance_status.
21	Food_Orders	Manages dining and room-service culinary orders. It cross-references back to a primary booking or a specific active room stay (room_booking_detail_id), tracking the ordering channel (order_type), real-time kitchen display status (kot_status - Kitchen Order Ticket), payment handling, and POS point-of-sale verification.
22	Food_Order_Details	A transactional line-item entity that captures the granular details of a food order. It documents exactly which culinary items were selected, the ordered quantity, and freezes the historical selling price (price_at_order) at the specific moment the ticket was sent to the kitchen.
23	Menu_Items	Houses the master catalog of all dishes, beverages, and culinary options available across the hotel's dining facilities. It keeps track of the official dish name, standard retail price, and an operational toggle (is_available) to easily handle out-of-stock menu options.
24	Folio_Items	Manages incidental charges, room service costs, and departmental expenses incurred by guests during their stay. It links back to an active booking segment, tracks the source department (e.g., Restaurant, Spa, Room Service), stores the calculated amount, and captures an optional employee signature (signature_img_url) for internal validation.
25	Consolidated_Invoices	Acts as the master billing statement for an entire customer booking. It aggregates all costs from room stays, tour

		reservations, and incidental folio items, handles fiscal calculations like tax (vat_amount) and pre-tax balances, applies active promotional codes, and monitors the legal document's issuance timeline.
26	Promotions	Stores active marketing and discount campaigns configuration within the system. It contains specific programmatic parameters including unique voucher alphanumeric strings (promo_code), discount types (e.g., percentage or fixed amount), validity thresholds (valid_to), and an operational availability toggle.
27	Payment_Transactions	Tracks distinct monetary funding actions processed against generated invoices. It captures precise accounting logs such as the processed transaction amount, chosen gateway methods (e.g., Credit Card, Cash, Digital Wallet), and the authoritative processing status to secure historical payment receipts.

2. Use Case Specifications

*[Provide specifications for the use cases (UCs) those are covered in the system. The UCs are grouped by the system features and even sub features. **You just need to provide UC specifications for complex UCs involving in the main workflows (business processes).** Other UCs (i.e CRUD or data-viewing UCs) are simple, and you just need to refer the descriptions in the Functional Requirement (part 3) below]]*

2.1.1 Register

ID and Name:	UC-1 Register		
Primary Actor:	Guest	Secondary Actors	Mail Server, Google OAuth
Description:	Allows a guest to create a new user account in the system by filling in required information or using a Google account. After registration, the system sends a verification email to activate the account.		
Trigger:	The guest clicks the "Sign Up" button on the homepage.		
Preconditions:	- The guest has not registered before. - The guest has access to a valid email. - The guest has not used an email that already exists in the system.		
Postconditions:	- A new inactive user account is created. - A verification email is sent to the user. - The account is pending activation.		
Normal Flow:	1. The guest opens the registration page. 2. Guest enters username, email, password, full name, gender, date of birth, and address. 3. Guest clicks "Sign Up". 4. System checks for required fields (MSG02). 5. System checks for email duplication (MSG05) 6. The system encrypts the password and stores the account (BR-02). 7. System displays success toast: "Registration successful" (MSG03).		

ID and Name:	UC-1 Register		
Primary Actor:	Guest	Secondary Actors	Mail Server, Google OAuth
Description:	Allows a guest to create a new user account in the system by filling in required information or using a Google account. After registration, the system sends a verification email to activate the account.		
Trigger:	The guest clicks the "Sign Up" button on the homepage.		
Alternative Flows:	AF1: Guest chooses "Sign in with Google" → Redirected to Google OAuth → Successful → Auto-register user and proceed to verification.		
Exceptions:	EX1: Required fields missing → MSG02. EX2: Email already in use → MSG05. EX3: Password too weak → MSG14.		
Priority:	High		
Frequency of Use:	Very frequent – every new student		
Business Rules:	BR-01, BR-02, BR-03, BR-105		
Other Information:	- Email verification is required before login. - Google OAuth is optional but supported.		
Assumptions:	- Google OAuth is correctly configured. - The mail server is operational. - The user has internet access and a valid email.		

2.1.2 Login

ID and Name:	UC-2 Login		
Primary Actor:	Customer/ Employee	Secondary Actors	None
Description:	Allows a registered user to securely log into the system by entering their registered email address and password. Upon successful authentication, the system grants access to the user's dashboard based on their role.		
Trigger:	The guest clicks the "Login" button on the homepage.		
Preconditions:	- The user has a registered account associated with an email address. - The account is active and not suspended or banned. - The user is not currently logged into the system.		
Postconditions:	- The system successfully authenticates the user and creates a secure login session. - The user is redirected to the appropriate dashboard based on their role.		

	- The user's status is updated to Online.
Normal Flow:	1. The user opens the Login page. 2. The system displays the login form. 3. The user enters their registered email address and password. 4. The user clicks the "Login" button. 5. The system validates that all required fields are provided and that the email format is valid. 6. The system verifies that the email exists in the database. 7. The system verifies that the account status is active. 8. The system compares the entered password with the encrypted password stored in the database. 9. The system checks whether the account is currently locked. 10. The system authenticates the user. 11. The system creates a secure login session. 12. The system determines the user's role and permissions. 13. The system updates the user's status to Online. 14. The system redirects the user to the appropriate dashboard. 15. The system displays the dashboard interface.
Alternative Flows:	AF1: At Step 10, the system requires additional OTP/Email verification for a new or unrecognized device. After successful verification, resume at Step 11. AF2: At Step 14, the user is redirected to a role-specific dashboard (Customer, Receptionist, F&B Staff, Housekeeping, Tour Guide, Admin, or Manager).
Exceptions:	
Priority:	High
Frequency of Use:	Very frequent
Business Rules:	BR-SYS-01 BR-SYS-02 BR-SYS-03
Other Information:	- Login activities may be recorded for security monitoring. - OTP verification may be required for suspicious login attempts. - Users are redirected to different dashboards according to their assigned roles.
Assumptions:	- The authentication service is available. - The database connection is operational. - The user has a valid registered account. - Email/OTP services are functioning normally.

2.1.3 Forgot Password

ID and Name:	UC-3 Forgot Password		
Primary Actor:	Guest/ Customer	Secondary Actors	Mail Server
Description:	Allows a customer to reset the account password by verifying their registered email address and creating a new password.		
Trigger:	The user clicks the "Forgot Password" link on the Login page.		
Preconditions:	- User has a registered account - User remember and can access the registered email address. - The mail server is available.		
Postconditions:	- The password is updated successfully. - Old password becomes invalid. - User can log in using the new password.		
Normal Flow:	1. Users open the Login page. 2. Users click "Forgot Password". 3. The system displays the password recovery form. 4. Users enter the registered email address. 5. The system validates the email format. 6. System checks whether the email exists in the system. 7. The system sends an OTP/Email Code to the registered email. 8. Users enter the received OTP/Email Code. 9. The system validates the OTP/Email Code. 10. Users enter a new password and confirm it. 11. The system encrypts the new password. 12. System updates the account password. 13. The system displays a message: "Password reset successful." 14. Users are redirected to the Login page.		
Alternative Flows:	AF1: Invalid OTP -> At Step 9 -> The entered OTP is incorrect -> System displays an error message -> Customer re-enters the OTP.		
Exceptions:			
Priority:	High		
Frequency of Use:	Frequent		
Business Rules:	BR-SYS-01, BR-SYS-02		
Other Information:	- Password recovery is performed through email verification. - The customer must remember the registered email address.		
Assumptions:	- The mail server is operational. - The customer has internet access. - The registered email account is accessible.		

2.1.4 Search Available Rooms

ID and Name:	UC-4 Search Available Rooms		
Primary Actor:	Guest / Customer	Secondary Actors	System
Description:	Allows guests or customers to search for available room categories based on check-in date, check-out date, number of guests, and room requirements.		
Trigger:	The user selects room search on the website.		
Preconditions:	- The system is operational. - Room categories are available in the system. - The user provides search criteria.		
Postconditions:	- Available room categories are displayed. - Room availability information is returned to the user.		
Normal Flow:	1. The user accesses the room search page. 2. The system displays search forms. 3. User enters: • Check-in date. • Check-out date. • Number of guests. 4. User clicks Search. 5. The system validates search criteria. 6. The system checks room availability. 7. System filters room categories that satisfy: • Capacity requirements. • Availability during the selected period. 8. The system calculates room prices based on current pricing rules. 9. System displays available room categories with: • Room category name. • Capacity. • Price. • Description. 10. User reviews search results.		
Alternative Flows:	AF1: User is logged in -> Customer accesses the member homepage -> Customer performs room search directly from the booking page.		
Exceptions:			
Priority:	High		
Frequency of Use:	Very Frequent		
Business Rules:	BR-FO-01		

Other Information:	- Guests can search rooms without logging in. - Booking requires authentication.
Assumptions:	- Room categories have been configured. - Pricing information is available. - Availability data is up to date.

2.1.5 Book Room

ID and Name:	UC-5 Book Room		
Primary Actor:	Customer	Secondary Actors	Payment Gateway (VNPay Sandbox)
Description:	Allows a customer to create a room booking, select room categories, specify stay dates, provide guest information, and make a deposit payment to reserve the booking.		
Trigger:	The customer clicks the "Book Room" button after searching available rooms.		
Preconditions:	- The customer is logged in. - Selected room categories are available. - Check-in and check-out dates are valid. - The customer account is active.		
Postconditions:	- A booking record is created. - Room booking details are stored. - Deposit payment is recorded. - Booking status becomes Confirmed.		
Normal Flow:	1. Customer searches available rooms. 2. The system displays available room categories. 3. The customer selects one or more room categories. 4. Customer enters: • Check-in date. • Check-out date. • Number of rooms. • Number of guests. • Special requests (optional). 5. The system calculates booking cost. 6. System displays booking summary. 7. Customer confirms booking information. 8. The system creates a Booking record with status Pending. 9. The system creates a Room_Booking record. 10. The system creates Room_Booking_Details records. 11. The system calculates the required deposit amount. 12. The system redirects customers to the VNPay payment page. 13. Customer completes deposit payment. 14. Payment Gateway returns successful transaction results.		

	15. System updates booking status to Confirmed. 16. The system displays booking confirmation.
Alternative Flows:	AF1: Multiple Room Booking -> Customer selects multiple rooms -> System creates multiple Room_Booking_Details records under the same booking. AF2: Payment Timeout → If the customer does not complete the deposit payment within 15 minutes → System automatically cancels the booking → Room is released back to inventory → MSG: "Booking expired due to payment timeout"
Exceptions:	
Priority:	Critical
Frequency of Use:	Very Frequent
Business Rules:	BR-FO-01, BR-FO-02
Other Information:	- One booking may contain multiple rooms. - Physical room assignment is performed later during Check-in by the Receptionist.
Assumptions:	- Payment Gateway is operational. - Room pricing has been configured. - Room availability information is accurate.

2.1.6 Cancel Booking

ID and Name:	UC-6 Cancel Booking		
Primary Actor:	Customer	Secondary Actors	System
Description:	Allows a customer to cancel a confirmed room booking before the check-in date. The system determines whether a refund is applicable according to the cancellation policy.		
Trigger:	The customer selects a booking and clicks " Cancel Booking ".		
Preconditions:	- The customer is logged in. - Booking exists. - Booking status is not Checked-In. - Booking status is not Cancelled.		
Postconditions:	- Booking status becomes Cancelled. - Refund transaction is created if eligible. - Room inventory becomes available again.		
Normal Flow:	1. Customer opens booking history. 2. The system displays customer bookings. 3. The customer selects a booking. 4. Customer clicks Cancel Booking.		

	5. The system displays cancellation information. 6. Customer confirms cancellation. 7. System checks cancellation policy. 8. The system calculates a refundable amount. 9. System updates booking status to Cancelled . 10. The system creates refund transactions if applicable. 11. The system displays cancellation results.
Alternative Flows:	AF1: Booking qualifies for full refund -> Cancellation occurs more than 48 hours before check-in -> System approves full refund.
Exceptions:	
Priority:	High
Frequency of Use:	Medium
Business Rules:	BR-FIN-02
Other Information:	- Refund records are stored in Payment_Transactions. - Refund transactions use transaction_type = Refund.
Assumptions:	- Payment records exist. - The booking belongs to the current customer. - Refund service is available.

2.1.7 Front Desk

ID and Name:	UC-7 Check-in		
Primary Actor:	Receptionist	Secondary Actors	Customer / System
Description:	Allows a receptionist to verify guest information, assign physical rooms, register accompanying guests, configure spending authorization, and complete the check-in process.		
Trigger:	The customer arrives at the resort and requests check-in.		
Preconditions:	- Booking status is Confirmed. - Deposit payment has been completed. - Assigned rooms are available. - Customer presents valid identification documents.		
Postconditions:	- Guest successfully checks in. - A physical room is assigned. - Booking status becomes Checked_In. - Room status becomes Occupied_Clean. - Guest spending authorization is configured.		
Normal Flow:	1. The receptionist opens the Check-in screen.		

	2. Receptionist searches for the booking. 3. The system displays booking information. 4. The receptionist verifies customer identity using CCCD or Passport. 5. The receptionist records information of accompanying guests. 6. The receptionist selects available physical rooms. 7. System assigns selected rooms to Room_Booking_Details. 8. Receptionist configures: • Spending authorization. • Credit limit. • Guest PIN. 9. Receptionist confirms check-in. 10. System updates Room_Booking_Details status to Checked_In. 11. System updates Booking status to Checked_In. 12. System updates Room status to Occupied_Clean. 13. The system completes the check-in process.
Alternative Flows:	AF1: Upgrade Dependent to Customer -> Receptionist selects "Upgrade Account" -> System creates a new Customer account for the dependent -> Existing dependent information is linked to the new account.
Exceptions:	
Priority:	Critical
Frequency of Use:	Very Frequent
Business Rules:	BR-FO-03, BR-FO-04
Other Information:	- Physical room assignment is performed during check-in. - Credit Limit and PIN are configured during check-in. - Dependents may be upgraded to Customer accounts.
Assumptions:	- Room inventory information is accurate. - The receptionist has proper authorization. - Customer identification data is valid.

ID and Name:	UC-8 Check-out		
Primary Actor:	Receptionist	Secondary Actor	Customer
Description:	Allows a receptionist to review all charges incurred during the guest's stay, generate a consolidated invoice, process final payment, and complete the check-out procedure.		
Trigger:	The customer requests check-out at the reception desk.		
Preconditions:	- Booking status is Checked_In. - A customer is currently staying at the resort. - All charges have been recorded in the system.		

Postconditions:	- Final payment is completed. - Consolidated invoice is issued. - Booking status becomes Checked_Out. - Room status becomes Vacant_Dirty. - The housekeeping task is automatically created.
Normal Flow:	1. The receptionist opens the Check-out screen. 2. Receptionist searches for the guest booking. 3. System retrieves booking information. 4. System retrieves all unpaid charges from Folio_Items. 5. System calculates: • Room charges. • Food & Beverage charges. • Tour charges. • Additional fees. 6. The system generates a consolidated invoice summary. 7. The receptionist reviews invoice details with the customer. 8. The receptionist verifies charge evidence if requested. 9. The system calculates outstanding balance. 10. The customer selects the payment method. 11. The receptionist processes payment. 12. System records payment transaction. 13. System issues consolidated invoice. 14. System updates Booking status to Checked_Out. 15. System updates Room status to Vacant_Dirty. 16. The system automatically creates a Housekeeping task. 17. System completes check-out.
Alternative Flows:	AF1: Apply Promotion -> Customer has a valid promotion code -> System applies the corresponding discount.
Exceptions:	
Priority:	Critical
Frequency of Use:	Very Frequent
Business Rules:	BR-FIN-01, BR-FO-04
Other Information:	- The invoice includes charges from Rooms, Food & Beverage, and Tours. - Payment transactions are stored in Payment_Transactions. - Housekeeping tasks are automatically generated after check-out.
Assumptions:	- All Folio_Items have been correctly recorded. - Payment services are available. - The receptionist has permission to perform check-out.

2.1.7 Restaurant POS

ID and Name:	UC-9 Place Food Order		
Primary Actor:	Customer	Secondary Actor	System/ F&B Staff
Description:	Allows a customer to select food and beverage items from the menu and submit an order for room service.		
Trigger:	The customer accesses the Room Service page and selects food items.		
Preconditions:	- The customer is currently checked in. - Customers have access to the Room Service page. - Selected menu items are available.		
Postconditions:	- A food order is created. - Order details are stored. - The kitchen receives the order. - Order status becomes Pending.		
Normal Flow:	1. The customer opens the Room Service page. 2. The system displays available menu items. 3. The customer selects food and beverage items. 4. The customer enters a quantity for each item. 5. The system calculates the order total. 6. The customer reviews the order. 7. The customer confirms the order. 8. The system verifies that selected items are available. 9. The system creates a Food_Order record. 10. The system creates Food_Order_Detail records. 11. The system sets order status to Pending. 12. The system sends the order to the Kitchen Dashboard. 13. The system displays order confirmation.		
Alternative Flows:	AF1: Multiple Menu Items -> At Step 3 -> Customer selects multiple menu items -> System creates multiple Food_Order_Detail records. AF2: Room Charge Payment -> At Step 7 -> Customer chooses to charge the order to the room -> System records the order for later settlement during Check-out.		
Exceptions:			
Priority:	High		
Frequency of Use:	Very Frequent		
Business Rules:	BR-FB-01, BR-FB-02		
Other Information:	- Orders are stored in Food_Orders and Food_Order_Details. - New orders are initially created with Pending status. - Kitchen staff cannot modify order prices.		

Assumptions:	- Menu information is up to date. - Kitchen Dashboard is operational. - Customers are associated with an active room booking.
--------------	---

ID and Name:	UC-10 Process Food Order		
Primary Actor:	F&B Staff	Secondary Actor	System
Description:	Allows kitchen staff to receive food orders from the POS or Room Service system, update preparation status, and complete food preparation.		
Trigger:	A new food order is received by the Kitchen Dashboard.		
Preconditions:	- A food order exists. - Order status is Pending. - F&B staff is logged in.		
Postconditions:	- Order status is updated. - Prepared food is ready for serving. - Order reaches Served status.		
Normal Flow:	1. F&BStaff opens the Kitchen Dashboard. 2. The system displays all Pending food orders. 3. F&BStaff selects an order. 4. F&BStaff reviews order details. 5. F&BStaff clicks Start Preparing. 6. System updates order status to Preparing. 7. F&BStaff prepares the food. 8. F&BStaff completes food preparation. 9. F&BStaff clicks Mark as Served. 10. System updates order status to Served. 11. The system records the completion time. 12. System updates the dashboard.		
Alternative Flows:	AF1: Multiple Orders Processing -> At Step 2 -> Multiple pending orders exist -> F&B Staff processes orders one by one. AF2: Out-of-Stock Item Report -> At Step 4 -> F&B Staff discovers insufficient ingredients -> F&B Staff reports the item as unavailable.		
Exceptions:			
Priority:	High		
Frequency of Use:	Very Frequent		
Business Rules:	BR-FB-02		

Other Information:	- F&B Staff cannot create orders. - F&B Staff cannot modify prices. - F&B Staff can only update order status. - Order lifecycle follows: Pending -> Preparing -> Served
Assumptions:	- Orders are correctly received from POS or Room Service. - Kitchen Dashboard is operational. - Ingredient availability is accurately managed.

ID and Name:	UC-11 Pay Food Order		
Primary Actor:	F&B Staff	Secondary Actor	Customer
Description:	Allows restaurant staff to process payment for a food order either by immediate payment or by charging the order to a guest room.		
Trigger:	The food order has been completed and is ready for payment processing.		
Preconditions:	- Food order exists. - Order status is Pending. - Kitchen staff is logged in.		
Postconditions:	- Payment information is recorded. - The order is marked as paid or charged to the room. - Financial records are updated.		
Normal Flow:	1. F&B Staff opens the payment screen. 2. System displays served food orders. 3. F&B Staff selects an order. 4. The system calculates the total amount. 5. F&B Staff asks the customer to choose a payment method. 6. The customer selects Pay Immediately. 7. F&B Staff records payment information. 8. The system marks the order as paid. 9. The system updates financial records. 10. The system displays payment success confirmation.		
Alternative Flows:	AF1: Charge To Room -> At Step 6 -> Customer chooses Charge To Room -> FB Staff enters room number -> System verifies room information -> System verifies spending authorization -> System verifies remaining credit limit -> Customer enters PIN or provides signature -> System creates a Folio_Item record -> System marks the order as charged to room.		
Exceptions:			
Priority:	High		
Frequency of Use:	Very Frequent		

Business Rules:	BR-FB-01
Other Information:	- Immediate payment is typically used for external restaurant customers. - Charge-to-room orders are settled during Check-out. - PIN or signature evidence is stored for dispute resolution.
Assumptions:	- The POS system is operational. - Room information is up to date. - Customer authorization data is available.

ID and Name:	UC-12 Book Tour		
Primary Actor:	Customer	Secondary Actor	System
Description:	Allows a customer to book a tour schedule, select the number of participants, and reserve available seats for a specific departure.		
Trigger:	The customer selects a tour and clicks Book Tour .		
Preconditions:	- The customer is logged in. - Tour schedule exists. - Tour schedule status is Open. - Seats are available.		
Postconditions:	- Tour booking is created. - Reserved seats are updated. - Booking status becomes Confirmed.		
Normal Flow:	1. The customer opens the Tour page. 2. The system displays available tours. 3. The customer selects a tour. 4. The system displays available tour schedules. 5. The customer selects a schedule. 6. The customer enters the number of participants. 7. The system checks seat availability. 8. System displays booking summary. 9. Customer confirms booking information. 10. The system creates a Booking record. 11. The system creates a Tour_Booking record. 12. System updates booked_seats in Tour_Schedules. 13. The system confirms the booking. 14. The system displays booking success information.		
Alternative Flows:	AF1: Group Tour Booking -> At Step 6 -> Customer books for multiple participants -> System records participant_count greater than 1.		
Exceptions:			

Priority:	High
Frequency of Use:	Very Frequent
Business Rules:	BR-TR-01
Other Information:	- A booking is linked to a specific Tour_Schedule. - Booked seats are tracked through booked_seats. - Attendance will be recorded later during the AI Face Scan process.
Assumptions:	- Tour schedules have been created by the Tour Manager. - Seat availability information is accurate. - Customer information is valid.

ID and Name:	UC-13 Cancel Tour Booking		
Primary Actor:	Customer	Secondary Actor	System, Receptionist
Description:	Allows a customer or receptionist to cancel an existing tour booking. The system validates cancellation conditions, updates tour slot availability, processes refund rules if applicable, records audit logs, and notifies the customer.		
Trigger:	The customer selects Cancel Tour Booking from their booking history, or requests a receptionist to cancel the booking.		
Preconditions:	- The customer is logged into the system. - The tour booking exists. - The booking belongs to the customer or is accessed by an authorized receptionist. - The tour has not started. - The customer has not been checked-in for the tour. - The booking has not already been cancelled.		
Postconditions:	- Tour booking is marked as Cancelled. - Tour slot availability is updated. - Refund request is generated if applicable. - Audit Log is recorded. - Cancellation confirmation is sent to the customer.		
Normal Flow:	1. The customer opens the My Tour Bookings page. 2. The system displays the list of the customer's tour bookings. 3. Customer selects a tour booking. 4. The system displays the booking details and cancellation policy. 5. Customer clicks Cancel Booking. 6. The system asks the customer to confirm the cancellation request. 7. Customer confirms the cancellation. 8. The system validates the booking information.		

	9. The system verifies that the booking status is eligible for cancellation. 10. The system verifies that the tour has not started. 11. System updates the booking status to Cancelled. 12. System releases the reserved tour slot(s). 13. The system processes the refund according to the cancellation policy, if applicable. 14. The system records the cancellation activity in the Audit Log. 15. The system sends a cancellation confirmation notification/email to the customer. 16. The system displays a cancellation success message.
Alternative Flows:	AF1: At Step 13, the booking is eligible for a refund. The system processes the refund and resumes at Step 14. AF2: At Step 13, the booking was charged to a room folio. The system adjusts the folio balance and resumes at Step 14. AF3: At Step 1, a receptionist cancels the booking on behalf of the customer. The system records the action and resumes at Step 13.
Exceptions:	E1: Booking not found. E2: Tour has already started. E3: Customer has already checked in. E4: Refund transaction failed. E5: Session timeout.
Priority:	High
Frequency of Use:	Occasional
Business Rules:	BR-TR-01 BR-TR-02 BR-SYS-04
Other Information:	- Cancellation confirmation is sent to the customer via email. - Cancelled bookings remain in the system for auditing and reporting purposes. - Refund transactions are processed through the integrated payment gateway (Stripe/VNPay). - Released tour slots become immediately available for new bookings. - All cancellation records can be reviewed by authorized administrators through the Audit Log system.
Assumptions:	- The customer has a valid and authenticated account. - The tour booking exists in the system. - The payment gateway (Stripe/VNPay) is available for refund processing. - Tour schedule and booking information are up to date.

	- The customer has permission to cancel only their own bookings.
--	--

ID and Name:	UC-14 Make Online Payment		
Primary Actor:	Customer	Secondary Actor	System/ Payment Gateway (VNPAY Sandbox)
Description:	Allows a customer to make an online payment for a booking deposit through the VNPAY payment gateway.		
Trigger:	The customer confirms a booking and chooses online payment.		
Preconditions:	- A booking has been created. - Booking status is Pending. - The deposit amount has been calculated. - Payment gateway is available.		
Postconditions:	- Payment transaction is recorded. - Booking status becomes Confirmed. - Deposit payment is successfully completed.		
Normal Flow:	1. Customer reviews booking information. 2. The system calculates the required deposit amount. 3. Customer chooses Online Payment. 4. The system creates a payment request. 5. The system redirects the customer to VNPAY. 6. The customer enters payment information. 7. The customer confirms the payment. 8. VNPAY processes the transaction. 9. VNPAY returns a successful payment result. 10. The system verifies the payment response. 11. System records a Payment_Transaction. 12. System updates booking status to Confirmed. 13. The system displays payment success information.		
Alternative Flows:	AF1: Partial Deposit Payment -> At Step 2 -> Booking requires only a deposit payment -> System calculates the deposit amount rather than the full booking value.		
Exceptions:			
Priority:	Critical		
Frequency of Use:	Frequent		
Business Rules:	BR-FO-02		

Other Information:	- VNPay Sandbox is used for payment processing. - Successful transactions are stored in Payment_Transactions. - Deposit payments are linked to bookings.
Assumptions:	- VNPay service is operational. - Internet connection is available. - Payment information is valid.

ID and Name:	UC-15 Submit Feedback		
Primary Actor:	Customer	Secondary Actor	System
Description:	Allows customers to submit ratings and feedback for completed services, including rooms, tours, and dining experiences.		
Trigger:	Customer selects the Submit Feedback option for a completed booking.		
Preconditions:	- The customer is logged into the system. - The related booking has been completed. - The booking belongs to the customer. - The feedback submission period has not expired.		
Postconditions:	- Feedback is successfully recorded in the system. - Rating score and review content are associated with the selected booking. - Feedback is available for viewing and moderation.		
Normal Flow:	1. The customer opens the booking history page. 2. System displays completed bookings eligible for review. 3. Customer selects a completed booking. 4. The system displays the feedback form. 5. The customer enters a rating score. 6. Customer enters review comments. 7. Customer clicks Submit Feedback. 8. The system validates the input data. 9. The system saves the feedback information. 10. The system links the feedback to the corresponding booking. 11. The system displays a confirmation message.		
Alternative Flows:	AF1: At Step 5, the customer submits only a rating without comments. The system records the rating and resumes at Step 9. AF2: At Step 5, the customer submits feedback for multiple service categories (room, tour, dining). The system records all feedback entries and resumes at Step 9.		

Exceptions:	
Priority:	Critical
Frequency of Use:	Frequent
Business Rules:	BR-TR-03 BR-TR-04
Other Information:	- Feedback may include both a rating score and review comments. - Submitted feedback can be displayed publicly on the resort website. - Reviews are subject to moderation according to resort policies. - Customers can submit feedback only for services they have actually used.
Assumptions:	- The customer is authenticated. - Booking completion information is accurate. - The review system is available. - The booking belongs to the customer submitting the feedback.

3. Functional Requirements

[Provide descriptions about the system's functions/screens. The functions/screens are grouped by the system features, and even sub-features if needed. For the screens, you need to provide the screen layouts (mock-up screens) and relevant specifications if needed]

3.1 Feature Name1

3.1.1 SubFeature Name1.1

3.1.1.1 Screen/Function Name1

[Content #1: UI layout (Mockup screen prototype)]

[Content #2: brief descriptions of the screen/function, mapped to the relevant use cases]

[Content #3: provide further descriptions for the screen's components/fields using table format below]

Field Name	Description
Field Name1	Field description: data type min/max length or value, initial data, etc.
Field Name2	...
Field Group-Name1	
Field Name3	...
Field Name4	...
Field Group-Name2	
...	...

3.1.1.2 Screen/Function Name2

...

3.1.2 SubFeature Name1.2

...

3.2 User Authentication

3.2.1 User Register

...

3.2.2 User Login

...

3.2.3 Password Reset

...

3.3 System Administration

3.3.1 Master Data

3.3.1.1 Setting List

Setting List

(1)

All Types

(2)

All Statuses

Enter keyword(s) to search

Search

[New Setting](#)

Id	Name	Type ^	Value	Priority	Status	Action
1	Manager	User Role		2	Active	Edit Deactivate
3	Admin	User Role		1	Inactive	Edit Activate
4	Teacher	User Role		4	Active	Edit Deactivate

<

1

2

...

9

10

>

(3)

This screen allows the Administrator to:

- View Setting List: view list of current master data.
- Filter Setting List: filter master data by data types, statuses
- Search Settings: enter keyword(s) to search master data by their names or values
- Sort Setting List: sort master data list (ascending, descending) by clicking column headers

On the screen, s/he can also

- Activate/Deactivate Setting: change status of a specific inactive/active master data
- Choose to go to the Setting Details screens for adding new or updating an existing master data by clicking the New Setting or Edit link.

Field Description

Field Name	Description
(1)	Initial values: all the active setting names with null or blank type Hover the mouse to show the field name: "Setting Type"
(2)	Initial values: All Statuses, Active, Inactive (default value "All Status") Hover the mouse to show the field name: "Setting Status"
(3)	The change-status action is Activate or Deactivate depending on the current status of the relevant setting (Inactive or Active, respectively).

3.3.1.2 Setting Details

Setting Details

Name*

Manager

Type

User Role

Value

Priority

2

Status

☐ Active ☒ Inactive

Description

Mô tả cho role giảng viên

Submit

This screen allows the Administrator to:

- Add New Setting: add new master data.
- Update Setting Details: update details of a specific master data

Field Description

Field Name	Description
Name	Data type: non-digit string, max length of 20 characters
Type	Initial data values: all active setting names (with null or blank type)
Value	Data type: any string, max length of 100 characters
Priority	Data type: a positive integer
Description	Data type: any string, max length of 200 characters

3.3.2 User Management

3.3.2.1 User List

...

3.3.2.2 User Details

...

4. Non-Functional Requirements

3.1 External Interfaces

[This section provides information to ensure that the system will communicate properly with users and with external hardware or software/system elements.]

3.2 Quality Attributes

[List all the required system characteristics (quality attributes) specification. Some of the possible attributes are provided with the guide/descriptions are mentioned here]

3.2.1 Usability

[This section includes all those requirements that affect usability. For example, specify the required training time for a normal user and a power user to become productive at particular operations]

specify measurable task times for typical tasks or base the new system's usability requirements on other systems that the users know and like specify requirement to conform to common usability standards, such as IBM's CUA standards Microsoft's GUI standards]

3.2.2 Performance

[The system's performance characteristics are outlined in this section. Include specific response times. Where applicable, reference related Use Cases by name.

Response time for a transaction (average, maximum)

Throughput, for example, transactions per second

Capacity, for example, the number of customers or transactions the system can accommodate

Resource utilization, such as memory, disk, communications, and so forth.]

3.2.3 ...

5. Requirement Appendix

[Provide business rules, common requirements, or other extra requirements information here]

5.1 Business Rules

[Provide common business rules that you must follow. The information can be provided in the table format as the sample below]

ID	Rule Definition
BR-SYS-01	All personal identification information (Citizen ID, Passport) and passwords must be encrypted before being stored in the database. Passwords use BCrypt hashing; identity documents use AES-256 encryption.
BR-SYS-02	Accounts that fail login more than five consecutive times shall be locked for 15 minutes. Email Codes expire after 3 minutes.
BR-SYS-03	Employee sessions shall automatically expire after 15 minutes of inactivity.
BR-SYS-04	All critical actions such as booking cancellation, invoice modification, permission changes, and pricing updates must be recorded in the Audit Log and cannot be deleted.
BR-SYS-05	Customer deletion requests must be handled using soft deletion by anonymizing personal information while preserving transaction records.
BR-FO-01	The system must prevent room overbooking by ensuring that the same physical room cannot be assigned to overlapping reservations. Database transactions using SELECT ... FOR UPDATE must be applied.
BR-FO-02	Online room bookings remain reserved for 15 minutes only. If no deposit payment is received within this period, the booking is automatically cancelled and the room is released back to inventory.
BR-FO-03	The representative guest performing check-in must be at least 18 years old and must present valid identification documents (Citizen ID or Passport).
BR-FO-04	Room status must follow the lifecycle: Vacant_Clean → Occupied → Vacant_Dirty → Vacant_Clean. Only rooms with Vacant_Clean status may be assigned during check-in.

BR-FO-05	Rooms flagged as Rush Room must be prioritized and moved to the top of the housekeeping task queue.
BR-FB-01	Food and room service charges may only be posted to rooms that are currently occupied and belong to the registered guest. PIN verification and Credit Limit checks are required before recording any Folio charge.
BR-FB-02	When a menu item is marked as unavailable by the kitchen, it must immediately become unavailable across all POS terminals and customer ordering interfaces via real-time broadcast.
BR-FB-03	Reserved restaurant tables are held for a maximum of 30 minutes after the scheduled reservation time. If the customer has not arrived, the table is automatically released.
BR-TR-01	The system must prevent tour overbooking by ensuring that the total number of booked participants cannot exceed the available seats of a Tour Schedule.
BR-TR-02	AI Face Scan attendance is considered valid only when facial similarity reaches at least 85% compared to the registered customer image. Results below this threshold must fall back to manual attendance.
BR-TR-03	Customers may submit reviews only within 7 days after tour completion or room check-out.
BR-TR-04	Administrators may hide or show reviews but are not permitted to modify review content. A moderation reason must always be recorded in the Audit Log.
BR-FIN-01	Check-out is not permitted until all outstanding balances in the consolidated invoice have been fully settled.
BR-FIN-02	Room cancellations made more than 48 hours before the scheduled check-in date receive a full deposit refund. Cancellations within 48 hours or no-shows receive no refund.
BR-FIN-03	The Night Audit process shall automatically run at 02:00 AM every day to close the previous business day, post nightly room charges to guest folios, and roll over to the new business date.
BR-FIN-04	Revenue reports must classify all income into three distinct categories: Room Revenue, Food & Beverage Revenue, and Tour Revenue, in accordance with USALI standards.
BR-FIN-05	A tour schedule is automatically cancelled and 100% refunds are issued to all registered participants if the total confirmed bookings fall below the minimum participant threshold at 24 hours prior to departure.

5.2 System Messages

#	Message code	Message Type	Context	Content
1	MSG01	In line	There is not any search result	<i>No search results.</i>
2	MSG02	In red, under the text box	Input-required fields are empty	<i>The * field is required.</i>

3	MSG03	Toast message	Updating asset(s) information successfully	<i>Update asset(s) successfully.</i>
4	MSG04	Toast message	Adding new asset successfully	<i>Add asset successfully.</i>
5	MSG05	Toast message	Confirming email of asset hand-over is sent successfully	<i>A confirmation email has been sent to {email_address}.</i>
6	MSG06	Toast message	Resetting asset information successfully	<i>Return asset(s) successfully.</i>
7	MSG07	Toast message	Deleting asset information successfully	<i>Delete asset(s) successfully.</i>
8	MSG08	In red, under the text box	Input value length > max length	<i>Exceed max length of {max_length}.</i>
9	MSG09	In line	Username or password is not correct when clicking sign-in	<i>Incorrect user name or password. Please check again.</i>
10	MSG10	In red	Login fields blank	<i>Email or password cannot be empty.</i>
11	MSG11	In line	Email not verified	<i>Account not verified. Please check your email for the verification link.</i>
12	MSG12	In line	Account inactive	<i>The account is blocked or inactive. Contact administrator.</i>
13	MSG13	Toast	Account locked	<i>Account locked for 15 minutes due to multiple failed login attempts.</i>
14	MSG14	In red	Weak password	<i>Passwords must be at least 8 characters, including letters, numbers, and symbols.</i>

5.3 Other Requirements...